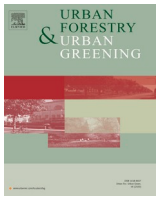
Urban Forestry & Urban Greening 118 (2026) 129362



Contents lists available at ScienceDirect

Urban Forestry & Urban Greening

journal homepage: www.elsevier.com/locate/ufug



Original article

Toward fully automated urban tree inventory: Integrating mobile lidar and open-source tools

Justinn J. Jones [a,*], Sorin C. Popescu [a], Joshuah S. Perkin [a], Stephen L. Webb [b]

[a] Department of Ecology & Conservation Biology, Texas A&M University, 2258 TAMU, College Station, TX 77843, USA
[b] Department of Rangeland, Wildlife & Fisheries Management, Texas A&M University, 2138 TAMU, College Station, TX 77843, USA

ARTICLE INFO

Keywords:
Mobile lidar scanning
Urban tree inventory
Tree biometrics
Open-source data science
Data visualization
Ecoinformatics

ABSTRACT

Urban trees provide numerous environmental benefits in built environments. To understand and measure ecosystem services provided by urban trees, we must be able to map, identify, and measure individual trees as accurately and efficiently as possible. Automating this approach using lidar and aerial photography can improve efficiencies and affordability of urban tree inventories. In addition to tree biometrics, identifying and measuring trees in urban areas is important for city planning, locating hazardous trees, and assessing tree influence on carbon and water cycling. The overall goal of this study was to map and model urban trees and estimate their biophysical parameters (i.e., tree height, diameter at breast height, and crown width), using aerial and mobile lidar with high-resolution aerial photography. The specific objectives were to: 1) develop the methodologies of mobile lidar data collection, 2) create a process to remotely measure trees from on-screen lidar point clouds to provide an efficient method of "ground truthing" data as a surrogate for field inventory, and 3) establish and incorporate processes to automate the tree measuring and validation workflow. Two open-source automated tree detection R packages were auditioned to automate the digital tree measuring process: lidR and TreeLS. The lidR package generated heights for large trees ($R^2 = 0.74$, RMSE = 1.07 m, p-value < 0.001, y = 0.90 x + 1.62) and shorter trees ($R^2 = 0.82$, RMSE = 0.45 m, p-value < 0.001, y = 1.02 x – 0.16). The TreeLS package generated dbh ($R^2 = 0.91$, RMSE = 5.25, p-value < 0.001, y = 1.06 x – 4.78). We demonstrated that mobile lidar data collection was an accurate and viable alternative to traditional forest inventory methods.

1. Introduction

1.1. Importance of urban trees

Urban trees provide numerous environmental benefits to the built environment, such as improved air and water quality, carbon sequestration, temperature regulation, and biodiversity (Nowak, 2006, Velasquez-Camacho et al., 2021). To better understand the importance of the various ecosystem services provided by urban trees, we must be able to map, identify, and measure individual trees as accurately and efficiently as possible. Urban tree inventories that estimate tree measurements can be resource intensive because there is often an emphasis on measuring individual trees (Wood, 1999).

Automating the measurement approach using lidar and aerial photography can improve efficiencies and affordability of urban tree inventories. Chen, S. et al. (2019) demonstrated time efficiencies of mobile lidar data collection compared to traditional field measurements; Ryding et al. (2015) reported time efficiency of mobile lidar over both terrestrial laser scanning and field measurements; Lindenbergh et al. (2015) emphasized computational efficiency of data-reduced, vehicle-based lidar for large-scale tree detection. Shrestha and Wynne (2012) found that high levels of accuracy are achieved for estimating tree height ($R^2 = 0.89$), diameter at breast height ($R^2 = 0.82$), crown diameter ($R^2 = 0.90$), and biomass ($R^2 = 0.67$) from airborne lidar data. Liu et al. (2017) determined that the combination of airborne lidar and hyperspectral imagery to classify common urban tree species yields an overall accuracy of 70.0 % compared to lidar (61.0 %) or hyperspectral (51.1 %) variables alone. Beyond basic tree biometrics, the identification and measurement of urban trees is critical for improved city planning, the detection of hazardous trees, and the evaluation of their role in climate regulation, carbon sequestration, and hydrological processes. Municipalities and urban planners could utilize mobile lidar data for

* Corresponding author.
E-mail addresses: justinn.j.jones@tamu.edu (J.J. Jones), Sorin.Popescu@ag.tamu.edu (S.C. Popescu), joshuah.perkin@ag.tamu.edu (J.S. Perkin), stephen.webb@ag.tamu.edu (S.L. Webb).

https://doi.org/10.1016/j.ufug.2026.129362
Received 19 March 2025; Received in revised form 10 February 2026; Accepted 19 February 2026
Available online 20 February 2026
1618-8667/© 2026 The Authors. Published by Elsevier GmbH. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

systematic monitoring, risk assessment, and the development of climate adaptation strategies.

1.2. Lidar systems: mobile vs terrestrial vs airborne

The term 'mobile lidar' is not standardized, and therefore can refer to different types of systems, depending on the context. Some studies, such as Herrero-Huerta et al. (2018), refer to mobile lidar as a large, multi-component system that is permanently mounted to a dedicated vehicle intended to collect data solely from roadways. We refer to mobile lidar scanning (MLS) to specifically mean a handheld scanner that can optionally, and temporarily, be mounted to a backpack, passenger vehicle, or UAV / UAS / drone (Fig. 1).

Due to its ground-based collection, we can also refer to this as terrestrial mobile lidar. This is to make the distinction from traditional terrestrial lidar scanning (TLS), which is mounted to a tripod and rotates to collect data 360° around the survey point. However, TLS only gives one vantage point of the target; to collect data on "the other side" of the target, the tripod must be moved to multiple locations, and the scans must be co-registered during post-processing.

Airborne lidar scanning (ALS) is collected from an aircraft, above a target, giving a bird's eye view looking down at the tree canopy. This allows for a detailed reconstruction of the top of canopy (TOC), in addition to penetration through the foliage; however, this vantage point provides minimal information about the tree stem, such as diameter at breast height (dbh). A benefit to ALS is that large geographic areas can be collected efficiently, but these data tend to be expensive to contract, so years may pass between collections. Contemporary ALS is of moderate spatial resolution. For example, the dataset used in this study has a posting (average spacing between points in the point cloud) of 0.4 m and a point density of 4.3 points/m^2.

Terrestrial MLS is collected from the ground, on foot or by vehicle, giving a man's eye view of the tree stem and canopy, allowing for a detailed reconstruction of both. A drawback to MLS is that there is a geographic limit to the efficiency of data collection: below which, MLS is ideal, above which, ALS is ideal; the precise geographic limit is dictated by many factors, including equipment and data collection methodologies. A technical limitation of any lidar collection is occlusion, or "feature shadowing" due to scanner vantage point. Of special concern to MLS collection is data drift, explained in greater detail in the discussion section. A major benefit of mobile lidar is the low relative cost of data collection, so MLS data can have a high temporal resolution. Spatial resolution can be viewed with ambivalence: high resolution is a plus, but the resulting large data files and increased processing time can be limiting factors.

1.3. Previous studies

Much work has been conducted on mapping and modeling trees with the aid of various remote sensing technologies, including imagery and lidar (Table 1). However, given the nature of mobile lidar compared to airborne or spaceborne lidar, the latter two sources will not be reviewed in detail here.

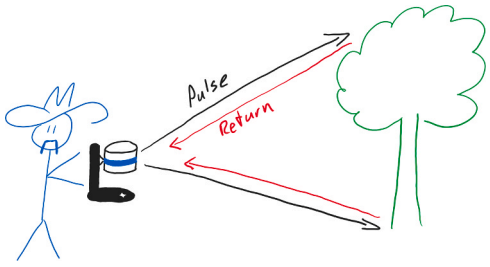


Fig. 1. Generalization of how handheld, mobile lidar scanning works. Light pulses are emitted by the device, and the light reflection (return) is recorded by the sensor. By precisely measuring the time between pulse and return, and considering the speed of light, distances can be calculated accurately.

Thorough reviews of related technologies and their applications in forestry have been conducted by Srinivasan et al. (2015), Safaie et al. (2021), Velasquez-Camacho et al. (2021), and Alvites et al. (2022). Of note, Alvites reviewed over 300 peer-reviewed papers to catalog recent lidar applications in forestry, including ALS, TLS, and spaceborne platforms, and Velasquez-Camacho reviewed 48 papers published 2016–2020 related to characterizing urban trees with various remote sensing technologies, including lidar and multispectral imagery.

Comparisons of MLS to TLS are detailed by Ryding et al. (2015), Bauwens et al. (2016), and Cabo et al. (2018), with Stal et al. (2021) going a step further to compare these systems with ALS. Bauwens reports a tree detection rate of 80 %, but due to the limited range of the scanner (15 m), the data were unusable for tree heights. Cabo compared a backpack laser scanner to TLS for measuring height and dbh. They showed favorable results of mobile lidar in estimating dbh measurements, but less favorable in height. Two major concerns in this study were that the scanner only had a range of 15 m, and sunny conditions were less than ideal for scanning. In comparing MLS to TLS, Ryding reported an MLS survey time of 50 m^2/min compared to TLS (0.85 m^2/min) and field data collection (0.43 m^2/min), highlighting the time efficiency of MLS.

Airborne laser scanning has been used extensively to estimate tree measurements. Popescu and Wynne (2004) describe a remarkably effective method for estimating tree height from ALS through the use of a variable window size and a local maximum filter on a lidar-derived canopy height model. Of note, this work was integral to the automation of tree height delineation step further described later. Popescu and Zhao (2008) used a variable window on airborne lidar data to locate 117 trees and measure height and crown width. Shrestha and Wynne (2012) showed great success in estimating tree dimensions from ALS with strong agreement with traditional field surveys, including height (R^2 = 0.89), dbh (R^2 = 0.82), crown diameter (R^2 = 0.90) and biomass (R^2 = 0.67). Dalponte and Coomes (2016) delineated individual tree crowns from ALS using a region-growing algorithm and were able to predict stem diameter based on the tree height / crown width relationship. Of note, this work is reflected in the lidR package (Roussel et al., 2020, Roussel and Auty, 2024) used to delineate crown areas in our study. Liu et al. (2017) determined that the combination of airborne lidar and hyperspectral imagery to classify common urban tree species yields an overall accuracy of 70.0 % compared to lidar (61.0 %) or hyperspectral (51.1 %) variables alone.

Terrestrial laser scanning can provide precise and accurate measurements about a target but can be cumbersome and time-inefficient due to being tripod-bound. Srinivasan et al. (2015) used TLS to measure height, dbh, and crown width of 380 trees in a research park and national forest. Their study found a height accuracy of 92 % with an RMSE = 1.51 m; dbh R^2 between 0.91 and 0.97; and crown width R^2 = 0.84 with RMSE = 1.08 m. Putman and Popescu (2018) used two different TLS systems to detect and characterize standing dead trees from three study sites in Texas.

Vehicle-based lidar scanning is shown to be useful in locating and measuring trees and other objects of interest in both urban and natural environments. Herrero-Huerta et al. (2018) used a Fugro Drivemap system that includes two laser scanners, GNSS receiver, and IMU, all permanently mounted to a dedicated vehicle to measure various biophysical parameters of urban trees in The Netherlands. The reported optimal estimation parameters give an R^2 = 0.92 and RMSE = 4.9 cm for dbh, and R^2 = 0.98 and RMSE = 16.2 cm for height. Wu et al. (2013) presented a new voxel-based approach to detecting, segmenting, and measuring urban trees from vehicle-based lidar data. Additionally, they incorporate a method for eliminating non-tree pole-like objects, such as streetlamps and road signs. These are promising results, but the large form factor of the whole system makes it unusable for urban trees that are not immediately adjacent to a road.

Table 1
Overview of relevant studies: urban environment (U), natural environment (N); airborne lidar scanning (ALS), tripod-mounted terrestrial lidar scanning (TLS), vehicle-mounted mobile lidar scanning (CAR), unmanned aerial vehicle / system (UAV), handheld / backpack-mounted mobile lidar scanning (MLS); number of trees (nt); parameters related to our study (Pars): height (h), diameter at breast height (dbh), crown diameter / crown width (cw).

Study	Env.	Scan	nt	Pars	Results	Notes
Aschoff and Spiecker, 2004	N	TLS	n/a	dbh	New algorithm for semi-automatic detection	Circle fitting
Bauwens et al., 2016	N	MLS/TLS	331	dbh	Detection: 80–90 %; dbh RMSE= 1.11–3.73 cm	MLS max range 15 m, unusable for height
Bienert et al., 2006	N	TLS	79	dbh	Mean diff. range: −0.93–0.93 cm; SD range: 1.21–2.47 cm	Circle fitting
Brolly and Kiraly, 2009	N	TLS	154	dbh	RMSE= 4.2 cm	Circle fitting (single)
Brolly and Kiraly, 2009	N	TLS	154	dbh	RMSE= 3.4 cm	Circle fitting (multiple)
Brolly and Kiraly, 2009	N	TLS	134	dbh	RMSE= 7.0 cm	Cylinder fitting
Cabo et al., 2018	U, N	MLS/TLS	171	h, dbh	h: RMSE= 1.3–9.4 m dbh: RMSE= 0.009–0.011 m	MLS max range 15 m, unusable for height
Chen, S. et al., 2019	N	MLS	30	dbh	Detection: 93.3 %; dbh: RMSE= 1.58 cm	nonlinear mixed models, 30 m range, time efficiencies
Chen, Y. et al., 2019	U	CAR	151	h, dbh	Detection: 93 %; dbh error: 5 %; dbh RMSE= 4.9–7.3 cm	RANSAC cylinder detection
Côté et al., 2018	N	TLS/ALS	n/a	h, dbh, cw	Reproduced forest structure at mean R^2 = 0.83	'L-Architect'algorithm, PlantGL library
Dalponte and Coomes, 2016	N	ALS	1762	h, dbh, cw	Detection varied with dbh class; h RMSE= 2.3 m; dbh RMSE= 11 cm	CHM local maxima filter for height
Fan et al., 2016	U	CAR	268	n/a	Detection: 86 %	Super-voxel segmentation
Forsman et al., 2016	N	CAR	257	dbh	dbh RMSE= 0.037–0.099 m	Circle fitting and trigonometric methods
Heo et al., 2019	U	MLS	54	h, dbh	h RMSE= 0.359–0.462 dbh RMSE= 3.77–8.95 cm	mobile lidar data proved incomplete for tree stem reconstruction
Herrero-Huerta et al., 2018	U	CAR	87	h, dbh	h RMSE= 16.2 cm, R^2 = 0.98 dbh RMSE= 4.9 cm, R^2 = 0.92	
Holopainen et al., 2013	U, N	ALS/TLS/CAR	438	n/a	Detection: 73.3–79.2 %	Focused on tree detection
Hopkinson et al., 2004	N	TLS	128	dbh	R^2 = 0.85	Cylinder fitting
Huang et al., 2009	N	TLS	26	dbh	R^2 = 0.79	Circle fitting
Husain and Vaishya, 2019	U	CAR	93	h, dbh, cw	Detection: 96.8 %	
Li et al., 2020	U	CAR	7	h, cw	Detection: 99 %	Discrete AdaBoost algorithm
Li et al., 2021	N	TLS	580	dbh		nonlinear mixed models
Monnier et al., 2012	U	CAR	41	n/a	RMSE: (h) 0.17 m, (dbh) 0.4 cm, (cw) 0.73 m	Probabilistic relaxation; RANSAC approach
Moorthy et al., 2011	N	TLS	24	h, cw	h: R^2 = 0.97, RMSE= 0.21 m cw: R^2 = 0.97, RMSE= 0.13 m	cross-sectional slicing
Moreira et al., 2021	N	UAV	14	dbh	dbh RMSE= 5 %-12.5 %	UAV imagery-based Structure from Motion
Popescu and Wynne, 2004	N	ALS	n/a	h	pine: R^2 0.97; deciduous: R^2 = 0.79	variable window, tree height
Popescu and Zhao, 2008	N	ALS	117	h, cw	h: R^2 = 0.95, RMSE= 1.55 m cw: R^2 = 0.53, RMSE= 1.84 m	
Rutzinger et al., 2010	U	CAR	40	h, dbh, cw	Detection: 93 %	
Ryding et al., 2015	N	MLS/TLS	49, 54	dbh	dbh: RMSE= 2.9 cm	MLS max range 15 m; Highlight time efficiencies in MLS over field collection
Saarinen et al., 2017	N	TLS	9	h, dbh	h: RMSE= 1.9–3.6 m	Cylinder fitting, Huber's formula
Safaie et al., 2021	U	CAR	15	h, dbh, cw	average differences: h= 8 cm, dbh= 2 cm, cw= 14 cm	Hough transformation
Shrestha and Wynne, 2012	U	ALS	3562	h, dbh, cw	h: R^2 = 0.89; dbh: R^2 = 0.82; cw: R^2 = 0.90	
Srinivasan et al., 2015	N	TLS	380	h, dbh, cw	h: RMSE= 1.51 m; dbh: R^2 = 0.91-0.97, cw: R^2 = 0.84, RMSE= 1.08 m	
Stal et al., 2021	N	MLS/TLS/ALS	25	dbh	dbh: RMSE= 0.024 m	MLS max range 15 m
Su et al., 2021	N	MLS	n/a	h, dbh	h: R^2 = 0.65, RMSE= 1.9 m dbh: R^2 = 0.95, RMSE= 0.02 m	
Tansey et al., 2009	N	TLS	8	dbh	RMSE range: 1.9–3.7 cm	Hough
Thies and Spiecker, 2004	N	TLS	26–11	dbh	dbh range: 82.3–109.5 %	Hough
Watt and Donoghue, 2005	N	TLS	12	dbh	R^2 = 0.92 (site 2)	Circle fitting
Wezyk et al., 2007	N	TLS	199	dbh	R^2 = 0.95	Cylinder fitting
Wu et al., 2013	U	CAR	72	h, dbh, cw	h: R^2 > 0.9; dbh: RMSE= 0.01–0.03 m, cw: R^2 > 0.9	Voxel-based Marked Neighborhood Searching (VMNS)

(*continued on next page*)

Table 1 (*continued*)

Study	Env.	Scan	nt	Pars	Results	Notes
Xu et al., 2018	U	CAR	172	n/a	Detection: 98.9 %	
Yan et al., 2019	U	CAR	30	h	Detection: 100 %	
Study	**Env.**	**Scan**	**nt**	**Pars**	**Results**	**Notes**
Yao et al., 2011	N	TLS	n/a	dbh	dbh R^2 = 0.34–0.85	ground-based, upward-scanning lidar; 'findtrunks'algorithm
Zhang et al., 2015	U	CAR	2652	h, cw	Detection: 93.5 %	
Zhong et al., 2013	U	CAR	5	h, dbh, cw	Detection: 100 %; h accuracy 0.1 m	
Zhong et al., 2017	U	TLS/ CAR	145–488	n/a	Accuracy= 94 %	hierarchical segmentation approach; modified node similarity calculation

Adapted from Srinivasan et al. (2015), Safaie et al. (2021), and Alvites et al. (2022).

Handheld mobile lidar scanning has gained popularity in ecological research. Chen, S. et al. (2019) used MLS to locate 30 trees with 93.3 % detection rate and measured dbh with an RMSE = 1.58 cm. Although they reported that the limited range of the scanner (30 m) was a hindrance to measuring tree heights, they were able to highlight the time efficiencies in MLS versus field measurements: 30 m^2/min compared to 0.91 m^2/min, respectively. Heo et al. (2019) used MLS to measure trees along a road (n = 15) and within a park (n = 39) in Seoul, Korea. Specifications for scan range and rate were not provided, but they report that their mobile lidar data proved incomplete for tree stem reconstruction. Su et al. (2021) showed high accuracies of dbh (R^2 = 0.95, RMSE = 0.02 m) and height (R^2 = 0.65, RMSE = 1.90 m) comparing field measurements to their custom-made backpack lidar scanner.

Some studies focused on tree (and other object) detection in general, rather than tree measurements specifically. Sirmacek and Lindenbergh (2015) developed an algorithm for classifying trees within ALS and vehicle-mounted lidar datasets. Windrim and Bryson (2020) present custom algorithms for tree detection, segmentation, and stem reconstruction from airborne lidar data. Golombek and Marshall (2021) were very successful at classifying urban streetscape features, such as trees, streetlamps, and traffic signs, from mobile lidar mounted to a vehicle. Measuring trees was not a major component of the study, other than to generally quantify them in a voxel context. This work can be useful in future study to classify and separate urban artifacts from target features, like urban trees, from a point cloud.

Tree species classification was investigated by Puttonen et al. (2011) and Wu et al. (2018) from vehicle-based lidar, as well as from airborne lidar by Wang et al. (2018). Liu et al. (2017) determined that the combination of airborne lidar and hyperspectral imagery to classify common urban tree species yields an overall accuracy of 70.0 % compared to lidar (61.0 %) or hyperspectral (51.1 %) variables alone.

Other studies explored topics adjacent to our investigations. Drones (unmanned aerial vehicles) are increasingly being used in conservation and ecological research (Wich and Piel, 2021). Moreira et al. (2021) measured dbh of olive trees in Spain using a Structure from Motion (SfM) point cloud derived from drone imagery. For the 14 trees in the study, they report dbh RMSE between 5 % and 12.5 %. Popescu et al. (2003), Popescu et al. (2004), and Popescu (2007) were highly successful at estimating tree volume and biomass from airborne lidar. These studies are integral for future investigations of volume, biomass, and carbon of urban trees with mobile lidar. In a study of 315 trees detected by vehicle-based lidar, Lindenbergh et al. (2015) focus on computational time efficiencies from voxel data reduction. Montoya et al. (2021) proposed an open-sourced tool, TreeTool, to measure dbh from detected trees within a point cloud. Their workflow shows promising results and may be useful in future study.

1.4. Novelty and relevance statement

To our knowledge, handheld / backpack-mounted mobile lidar scanning had not been collected or analyzed in our same manner as a tool for urban tree ecology. We developed a complete and accessible workflow for contemporary urban forestry. We collected mobile lidar data with a 100 m range scanner, resulting in accurate representations of tree canopies. We emphasized the importance of collecting mobile lidar scans 1) via scooter rather than by vehicle and 2) concurrent to GPS reference data collection. This electric scooter workflow was a targeted solution to the middle-scale gap in urban forestry: areas too extensive for walking surveys yet inaccessible to vehicle-mounted surveys. We developed the methodology of using on-screen mobile lidar measurements to 1) serve as a secondary validation tier of field measurements and 2) act as a high-fidelity surrogate for field data to develop and expand parameter extraction algorithms. We developed the Synthetic Point from Raster tool as a novel approach for lidar point thinning, resulting in significantly smaller datasets and shorter computational times. This adaptable solution directly addresses a "big data" problem in mobile lidar, with applications far beyond urban forestry. We explored and discussed algorithmic behavior on various age classes of trees, leading to a greater understanding and implementation of these processes. We identified algorithmic processing challenges and limitations, providing a practical assessment of the performance of forestry-centric tools in a complex urban environment. We employed traditional forest inventory methods to measure 122 urban trees for use as field validation. These novel improvements directly answered unresolved challenges from previous studies. The integrated workflow system, the rigor of multi-tiered validation, and the insights gained from algorithmic challenges were presented as significant methodological contributions. Our investigation was part of a growing movement toward semi-automation and scalability in urban forestry and urban remote sensing. We were able to demonstrate that mobile lidar data collection is an accurate and viable alternative to traditional forest inventory methods.

1.5. Goal and objectives

The overall goal of this study was to map and model urban trees and estimate common biophysical parameters (total tree height (h), diameter at breast height (dbh), and crown width (cw)), using aerial and mobile lidar with high-resolution aerial photography. The specific objectives were to: 1) develop the methodologies of mobile lidar data collection, 2) create a process to remotely measure trees from on-screen lidar point clouds to provide an alternative method of "ground truthing" data as a surrogate for field inventory, and 3) establish and incorporate processes to automate the tree measuring and validation workflow to the greatest extent possible with open-source tools. This study addresses questions such as: 1) Can mobile lidar achieve accurate results comparable to traditional, field-based measurements? 2) How well do open-source packages developed for trees in forest environments perform in urban tree settings? 3) How can urban forest inventory benefit from mobile lidar scanning? 4) Can mobile lidar datasets be processed in a semi-automated fashion to make the data accessible to a broader audience?

2. Methods & materials

2.1. Study Site

The study area was the Texas A&M University campus, College Station, Texas (30°36'19.81" N, 96°20'29.11" W) (Fig. 2). MLS data collections were conducted around the Leach Teaching Gardens on Carder Lane, down to The Bush School on Research Parkway, along the West Campus Garage on John Kimbrough Boulevard, around Simpson Drill Field, along Houston Street, and up to the Chemistry Building on Ross Street. The site was selected because it was representative of the local urban landscape by a mix of multistory buildings, parking lots and garages, walkways, and manicured lawns, shrubs, and trees. We believe that this mix of isolated trees, canopy corridors, and varied infrastructure provided adequate 3D structure to develop our methodologies within a controlled, heterogeneous environment that allowed for the isolation of algorithmic performance variables without the confounding factors of extreme understory density found in wildland forests.

2.2. Data

ALS data were available from the USGS 3D Elevation Program (USGS, 2023); the most recent vintage for our project area was from 2017. This dataset had a posting (average spacing between points in the point cloud) of 0.4 m and a point density of 4.3 points/m^2 (ppm) and was used to validate horizontal and vertical positional accuracy of our MLS.

Aerial photography from June 2022 (60 cm spatial resolution, 4-band (R/G/B/NIR) spectral resolution) was available from the National Agriculture Imagery Program (NAIP) (USDA, 2023) and was downloaded via the Texas Geographic Information Office (https://tnris.org). This imagery dataset was used as an input layer to spatially subset the MLS data.

Mobile lidar scans were conducted with a GeoSLAM ZEB Horizon

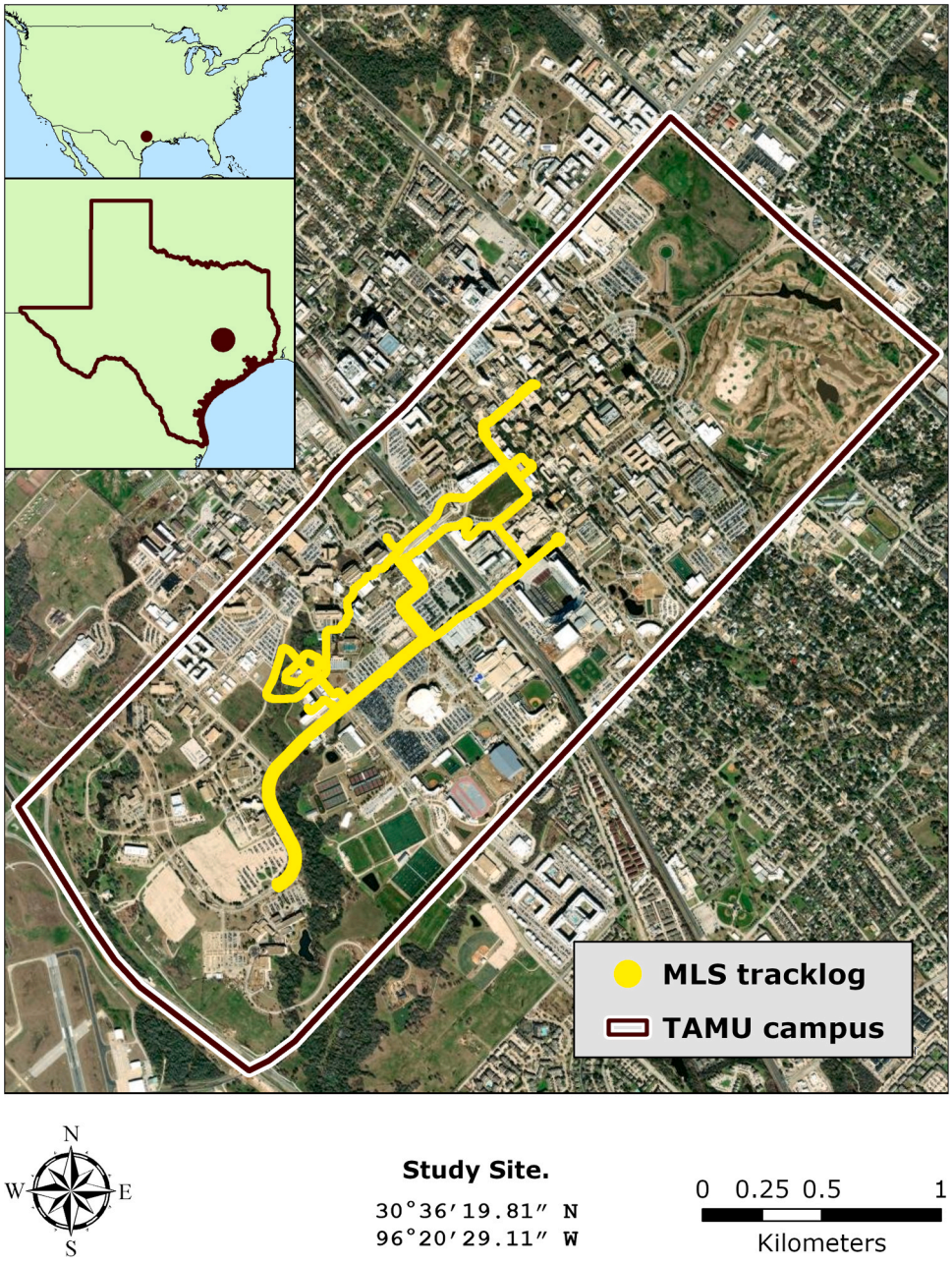


Fig. 2. Study site on Texas A&M University Campus, College Station, Texas, USA.

mobile lidar scanner mounted to a backpack system (Fig. 3). The scanner has a maximum range of 100 m, scan angle of 360° x 270° (as to exclude the operator), scan rate of 300,000 points/s, operates at 903 nm wavelength (in the near-infrared portion of the electro-magnetic spectrum), and weighs under 1.5 kg. The backpack was worn while riding a VEO electric scooter to capture trees from both roadways and pedestrian walkways at an average speed of 20 km/h. Data collected by vehicle is limited to the immediate area surrounding a road; a scooter allowed us to collect data without this limitation. As opposed to a pedal bicycle, an electric scooter can be operated at constant speed, for long periods of time, with little effort from the user. Total scan distance was 10.9 km. In testing, we found that multiple, short-duration scans (7–10 min each) yielded better results than fewer, long-duration scans. The total collection was split into 5 individual scans segments. These datasets were split, then merged in a way that no single area contained overlapping scans. The final lidar dataset had a posting of 0.04 m and a point density of 600 ppm (single pass, only including trees and ground, not buildings). Fig. 4 shows an example of the MLS dataset, and a sample is provided at the author's GitHub (JJJ GitHub: https://github.com/justinn-j-jones/research-projects). The full dataset is not provided due to size limitations.

The GPS reference data were collected concurrent to MLS on a Trimble Geo7x using the TerraSync Centimeter Edition application, with the aid of a Zephyr Model 2 external antenna mounted to the MLS backpack system (Fig. 3). By mounting the external antenna next to the MLS, we were able to georeference the MLS point cloud dataset efficiently and accurately. The alternative to this method would be to establish ground control prior to MLS collection, drive to each ground control point, dismount from the scooter, and detach the scanner from the backpack to physically occupy each survey location.

Tree field measurements were collected for ground-truth validation: height was measured using the height routine on a TruPulse 200 laser rangefinder, and dbh was measured with a Spencer steel diameter tape; a diagram of these biophysical parameters is shown in Fig. 5. Each tree was mapped using GPS equipment with positions averaged for 30 s. Initially, 50 trees from the south-end of the project area were measured; later, another 72 trees from the remaining project area were measured for a total of 122 field-measured trees. Trees were selected in the field at random, with a minimum height of 4 m, unobstructed view of the stem, and no fork at or below breast height. Although species was not considered when selecting individuals to sample, many sampled trees were live oak (*Quercus virginiana*) due to their abundance within the project area. Other common species in the project area include post oak (*Q. stellata*), bur oak (*Q. macrocarpa*), and Mexican white oak (*Q. polymorpha*). Sampled trees were identified to the genus or species level by the author. Collected tree heights ranged from 4.5 m to 18.2 m, with median = 8.8 m, mean = 9.3 m, and standard deviation = 3.04 m (Fig. 6). Diameters at breast height ranged from 8.9 cm to 99 cm, with median = 39.1 cm, mean = 42.8 cm, and standard deviation = 20.24 cm (Fig. 7).



Fig. 3. Mobile lidar scanner and GPS backpack system.

2.3. Data and acquisition costs

Both the 2017 USGS 3DEP ALS and the 2022 aerial photography are available free-of-charge from the USGS and Texas Geographic Information Office (https://tnris.org), respectively. The MLS and GPS data collection and processing was conducted with equipment and software owned by The LASERS Lab (https://lasers.tamu.edu) in the Department of Ecology and Conservation Biology at Texas A&M University. VEO electric scooter rental cost less than $30 (USD, 2023) for all necessary scans.

2.4. Processing approach

Data were processed on a Dell Precision Tower 7910 workstation running a 64-bit version of Windows10 Enterprise with an Intel Xeon E5–2643 CPU, NVIDIA Quadro M4000 GPU, and 256 GB of RAM.

2.5. Geospatial analysis

Airborne lidar scans were acquired as an LAS/LAZ file format and pre-processed in Quick Terrain Modeler v8.2.3 (QTM) (Applied Imagery, 2024) to remove noise, clip to the project area boundary, convert mean sea level (MSL) heights to above ground level (AGL) heights, and generate a canopy height model (CHM) at 0.5 m grid-sampling resolution to best represent the data. The clip tool was used to manually remove obvious noise from the point cloud, i.e., points that deviated significantly from expected spatial patterns of local trees and structures. For example, if there were scattered points more than 3 m above the top of tree canopy, these were considered noise and removed.

The QTM parameters used to mitigate noise when generating the CHM were: remove spikes (1 m) and aggressive level 3. An explanation of these parameters can be found in the product documentation.

Aerial photography was acquired as a JPG2000 and pre-processed within ArcGIS Pro v3.1.0 (AGP) (ESRI, 2024) to clip to the project area boundary and generate a Normalized Difference Vegetation Index (NDVI) to aid in isolating trees from non-target features such as road signs and streetlamps (Fig. 8).

GPS reference data were pre-processed in GPS Pathfinder Office v5.60 (Trimble, 2013) for location accuracy enhancement using a Continuously Operating Reference Station (CORS) from the Texas Department of Transportation station in Bryan, Texas. This dataset was processed in AGP to project the dataset to WGS 1984, reproject to UTM NAD83, calculate the geometry of X, Y, and Z, then generate a table of the UTM locations to georeference the mobile lidar scans. The final data formats were an ESRI shapefile (SHP) and an Excel spreadsheet. The Excel spreadsheet was converted to a TXT text file for later use.

Mobile lidar scans were pre-processed in GeoSLAM Hub v6.1.0 (FARO, 2024) and georeferenced to ground control using the GPS TXT file. The LAS dataset was further pre-processed in QTM to remove noise, clip to the project area boundary, convert MSL heights to AGL heights, and generate a CHM at 0.5 m grid-sampling resolution to best represent the data (Fig. 9). The clip tool was used to manually remove obvious noise from the point cloud, i.e., points that deviated significantly from expected spatial patterns of local trees and structures. The QTM parameters used to mitigate noise when generating the CHM were: remove spikes (1 m) and aggressive level 3. An explanation of these parameters can be found in the product documentation.

To validate that our MLS dataset had been georeferenced correctly, it was compared against the ALS dataset. An example of this process is shown in Fig. 5, where the green points represent the 2017 ALS, and red points represent 2023 MLS. This figure also shows that there is high agreement in the horizontal positioning of the MLS dataset compared to ALS. Furthermore, growth from 2017 to 2023 can be measured as the difference between these heights and widths.

In order to correlate field measurements to lidar-derived measurements, GPS locations were used. After the "digital trees" were identified

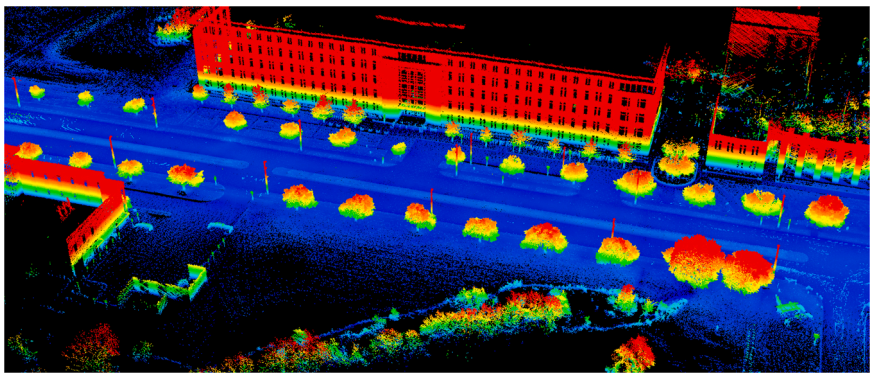


Fig. 4. Sample from mobile lidar scan showing urban trees. Points are colored by height: blue to red, low to high.

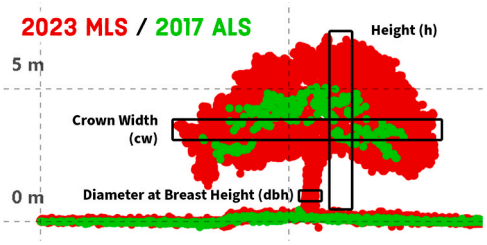


Fig. 5. Profile analysis in Quick Terrain Modeler showing standard forest tree biophysical parameters (h, dbh, cw). 2023 MLS points are shown in red, 2017 ALS points are shown in green.

within the lidar point cloud, we used a mensuration line and profile analysis tool in QTM to measure h, dbh, and cw; these are referred to as "screen" measurements (Fig. 10). Screen measurements were only collected on the initial 50 field-collected trees as proof of concept and source of further validation.

To automate the digital tree measuring process, two R packages were auditioned: lidR and TreeLS (Roussel et al., 2020, Roussel and Auty, 2024, De Conto, 2020). These specific packages were selected not only for their ability to yield the necessary tree measurements with customizable parameterization, but also for their open-source nature and establishment within contemporary forestry. Their application in a complex urban environment represents a novel aspect of our study. R version 4.3.1 (R Core Team, 2023) was run within RStudio version 2023.12.0 (RStudio Team, 2024). In addition to lidR and TreeLS, the R packages raster (Hijmans, 2023), sf (Pebesma, 2018, Pebesma and Bivand, 2023), ggplot2 (Wickham, 2016), terra (Hijmans, 2024), and mapview (Appelhans et al., 2023) were used in the investigation.

The lidR package was developed for ALS datasets with moderate posting and point density, like the ALS in this study. The 'locate_trees' function uses a local maximum filter on the point cloud to find a tree top as the local maximum Z value. The window size for the filter can be set to a specified width (acceptable for a homogeneous stand) or a variable window, where the window size increases in width with an increase in height by a user-defined function. This concept was based on the work pioneered by Popescu and Wynne (2004). The first 50 trees in the study were measured on screen for height, dbh, and crown width. This height to crown width ratio yielded an allometric equation ($y = 1.15x - 0.58$) that we could use as an input parameter into the variable window

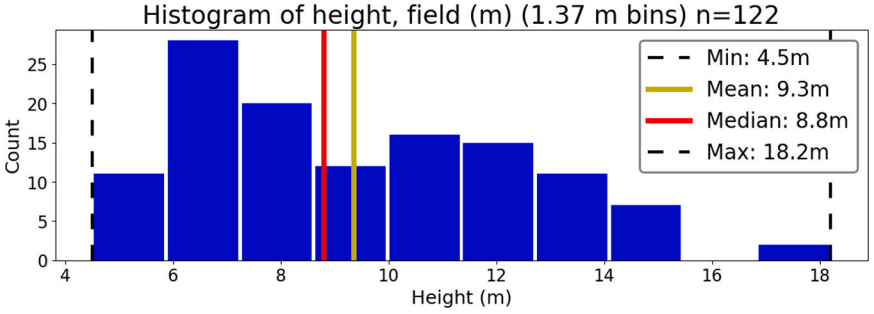


Fig. 6. Histogram of height for field sampled trees.

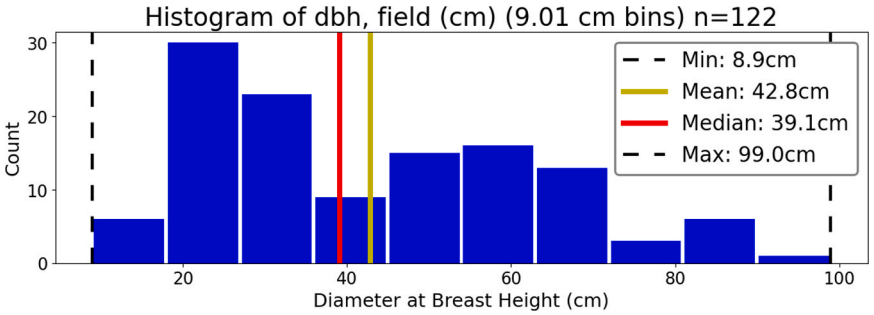


Fig. 7. Histogram of diameter at breast height for field sampled trees.

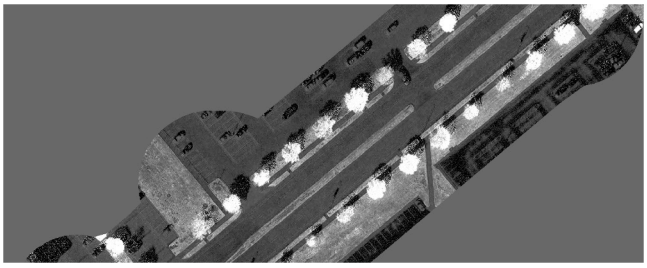


Fig. 8. An NDVI produced from 2022 4-band NAIP imagery, colorized black to white: low to high values. Grass is shown in light gray and trees are shown in white.

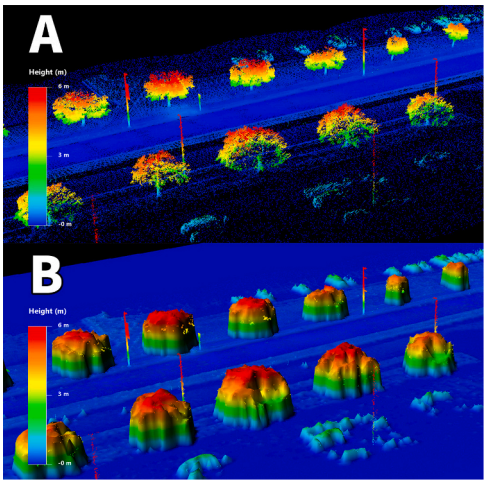


Fig. 9. Mobile lidar scans (A) were used to generate the canopy height model (B).

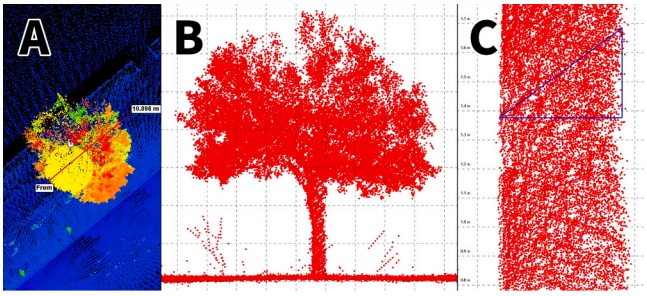


Fig. 10. Within QTM, a mensuration line is drawn over a tree in the point cloud (A), displayed as a profile (B), and measured for diameter at breast height (C).

function.

Because only the TOC points are needed in this function, and because MLS has a much higher point density than ALS, we elected to increase computational efficiency by performing a point resampling on the TOC using a custom tool: Synthetic Points from Raster (SPR). This, and all other programming notebooks, can be found at the author's GitHub (JJJ GitHub). This tool takes a raster as an input (the CHM in this case) and generates a single point per cell using the Z value. The output is a TXT with XYZ values. This TXT can be read into QTM as XYZ (ASCII), and an LAZ can be exported. Fig. 11 shows the original MLS point cloud, the TOC point cloud, the CHM raster, and a profile analysis in QTM. This process resulted in a point reduction to less than 5 % of the original (57.9 million to 2.5 million points).

In the 'segment_trees' function of the lidR package, the tree top points were fed into a seed-plus-growing-region algorithm on the CHM

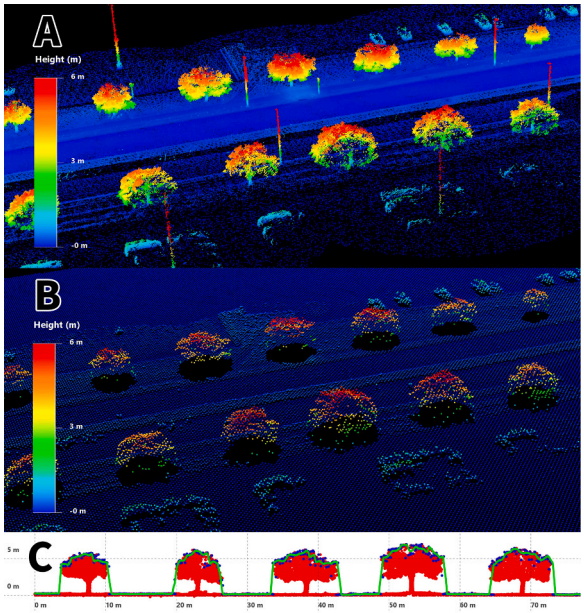


Fig. 11. Original MLS (A), top of canopy (TOC) points (B), and profile analysis in QTM (C): MLS points in red, TOC points in blue, and CHM raster in green. This process resulted in a point reduction to less than 5 % of the original (57.9 million to 2.5 million points).

to delineate crown areas. A detailed explanation can be found in the package documentation. This process (JJJ GitHub) resulted in an SHP for each parameter: points for tree tops and polygons for crown areas (Fig. 12).

These points and polygons were further evaluated within an interactive python notebook (IPYNB) using arcpy (ESRI, 2024), numpy (Harris et al., 2020), pandas (McKinney, 2010), matplotlib (Hunter, 2007), scikit-learn (Pedregosa et al., 2011), scipy (Virtanen et al., 2020), statsmodels (Seabold and Perktold, 2010), math (Van Rossum, 2020) and OS (Van Rossum, 2020) in order to tidy the data. Python version 3.9.18 (Python Software Foundation, 2023) was run within Visual Studio Code version 1.89.1 (Visual Studio Code, 2023).

To correlate field measurements with algorithm measurements, the nearest neighbor tool in APG was used to ensure that no pairs were outside of an acceptable distance, in this case, < 4 m because no two field-measured trees were within 4 m of each other. The pairs were spatially joined using the join tool in AGP. For the crown polygons, to convert crown area to crown width, 'Minimum boundary geometry' in AGP was used to calculate length and width of each polygon. These measurements were averaged to calculate cw_average. To improve upon this method, the equation for area of a circle was used to calculate cw_pi. These two calculations were averaged to calculate the improved crown width (icw). The processes in the notebook (JJJ GitHub) resulted in a CSV with locations of each tree and measurements: dbh and height (field); dbh, height, and crown width (screen); height, crown area, and improved crown width.

Within TreeLS (JJJ GitHub), tree stems were identified in the lidar point cloud by finding circles within specified height parameters. A cylinder was fit at 1.3 m, from which, dbh was measured. Additionally, tree height is measured from this location up to a specified percentile, in this case, a 0.999 height percentile parameter (99.9th percentile) to capture the tree top with minimal noise. A detailed explanation can be found in the package documentation. This process resulted in a CSV with tree locations, height, and dbh (Figs. 13 and 14). Again, the nearest neighbor tool was used to correlate field measurements to algorithm measurements.

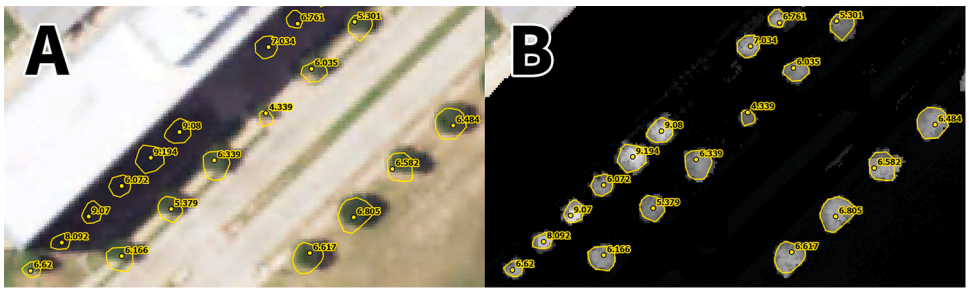


Fig. 12. Tree top points and crown area polygons generated from lidR code displayed on aerial photography (A) and CHM (B).

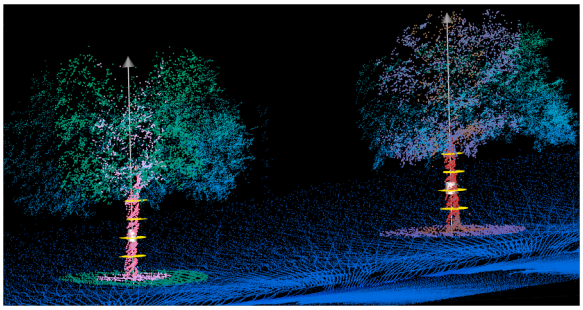


Fig. 13. Height and dbh are measured in the TreeLS code.

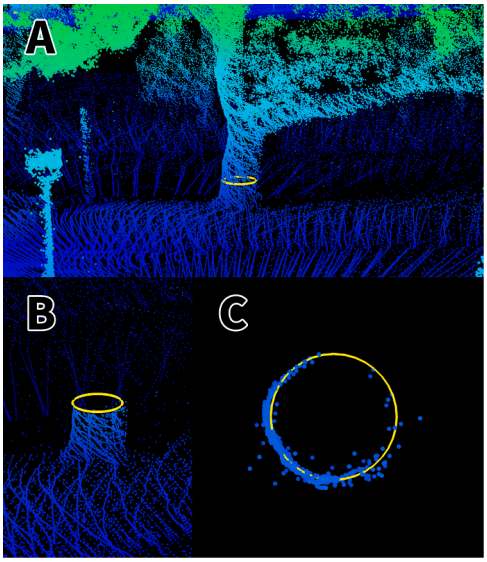


Fig. 14. TreeLS code dbh measurement shown within the point cloud in QTM (A); the digital tree is "cut down" to 1.3 m (B); top-down view of points at breast height and dbh ring (C).

2.6. Statistical analysis

To evaluate screen and code measurements against field data, paired tests were conducted. The paired *t*-test was appropriate when the difference between pairs followed a normal distribution, with the Wilcoxon signed-rank test representing the non-parametric, robust alternative. The null hypothesis for each of these tests was that the mean difference between the pairs was zero, in other words, that the upper and lower confidence intervals included zero. Normality of paired differences was evaluated with a Shapiro-Wilk test. The absolute value of the mean difference between pairs was also calculated and reported. To aid in the interpretation of results, effect size of the paired samples was evaluated using Cohen's d.

Next, a test for global validation of linear model assumptions was performed from the R package gvlma (Peña and Slate, 2006). This package generated a global statistic based upon four criteria: skewness, kurtosis, link function, and heteroscedasticity, with the null hypothesis being that the assumptions are acceptable, i.e., no violations to the assumptions of linear regression had been made. Finally, pairs of measurements were evaluated through simple linear regression and their respective plots were generated. A visual representation of data collection and processing methods can be found in Fig. 15.

3. Results

To assess the effectiveness of the various measurement methods, each was evaluated against field data. Paired test results are shown in Table 2. A Shapiro-Wilk (S-W) test for normality on the difference between measurement pairs yielded p-values greater than alpha (**$\alpha = 0.01$**), which dictated that we failed to reject the null hypothesis that the data were normally distributed and proceed with paired *t*-tests (PTT). PTT for each measurement pair, save field height vs lidR, yielded a p-value less than the alpha level, which indicated that the pairs were significantly different. The absolute value of the mean difference between pairs of height measurements ranged from 0.208 m (field vs lidR) to 1.600 m (field vs TreeLS). Mean differences of dbh ranged from 1.783 cm (field vs TreeLS) and 5.522 cm (field vs screen). Cohen's d (Cd) values of effect size for height measurements ranged from −1.417 (field vs TreeLS) to −0.238 (field vs lidR); dbh values ranged from 0.337 (field vs TreeLS) to 1.617 (field vs screen).

From the test results for global validation of linear model assumptions on the measurement pairs (Table 3), all but one pair met the assumptions. The pair for height: field vs lidR did not meet the assumptions due to heteroscedasticity and link function; transformations of log, log10, reciprocal, cube root, and square root did not change the spread of data in any meaningful way. Further examination of the field height histogram (Fig. 6) revealed a natural break in the data around 9 m, which likely distinguishes the population of younger, recently planted street trees from more mature, established individuals. The median value of 8.8 m was chosen because it 1) approximated the valley between the bimodal peaks of the histogram, and 2) split the dataset roughly in half, leaving 59 observations below 8.8 m and 63 observations at or above 8.8 m. Each of the above and below dataset pairs satisfied the linear model assumptions.

In regressing heights collected from field and screen (Fig. 16), we found $R^2 = 0.96$, RMSE = 0.52 m, p-value < 0.001, and $y = 1.06\ x - 0.14$. Comparing heights from field and lidR code, the regression analysis of trees at or above 8.8 m gave us an $R^2 = 0.74$, RMSE = 1.07 m, p-value < 0.001, and $y = 0.90\ x + 1.62$. The regression analysis of trees below 8.8 m gave us an $R^2 = 0.82$, RMSE = 0.45 m, p-value < 0.001, and $y = 1.02\ x - 0.16$. Comparing heights from field and TreeLS code, we found $R^2 = 0.82$, RMSE = 1.13 m, p-value < 0.001, and $y = 1.08\ x + 0.96$. In comparing dbh from field and screen (Fig. 17), the regression analysis gave us an $R^2 = 0.96$, RMSE = 3.39 cm, p-value < 0.001, and $y = 0.96\ x - 4.17$. Comparing dbh from field and TreeLS code, we

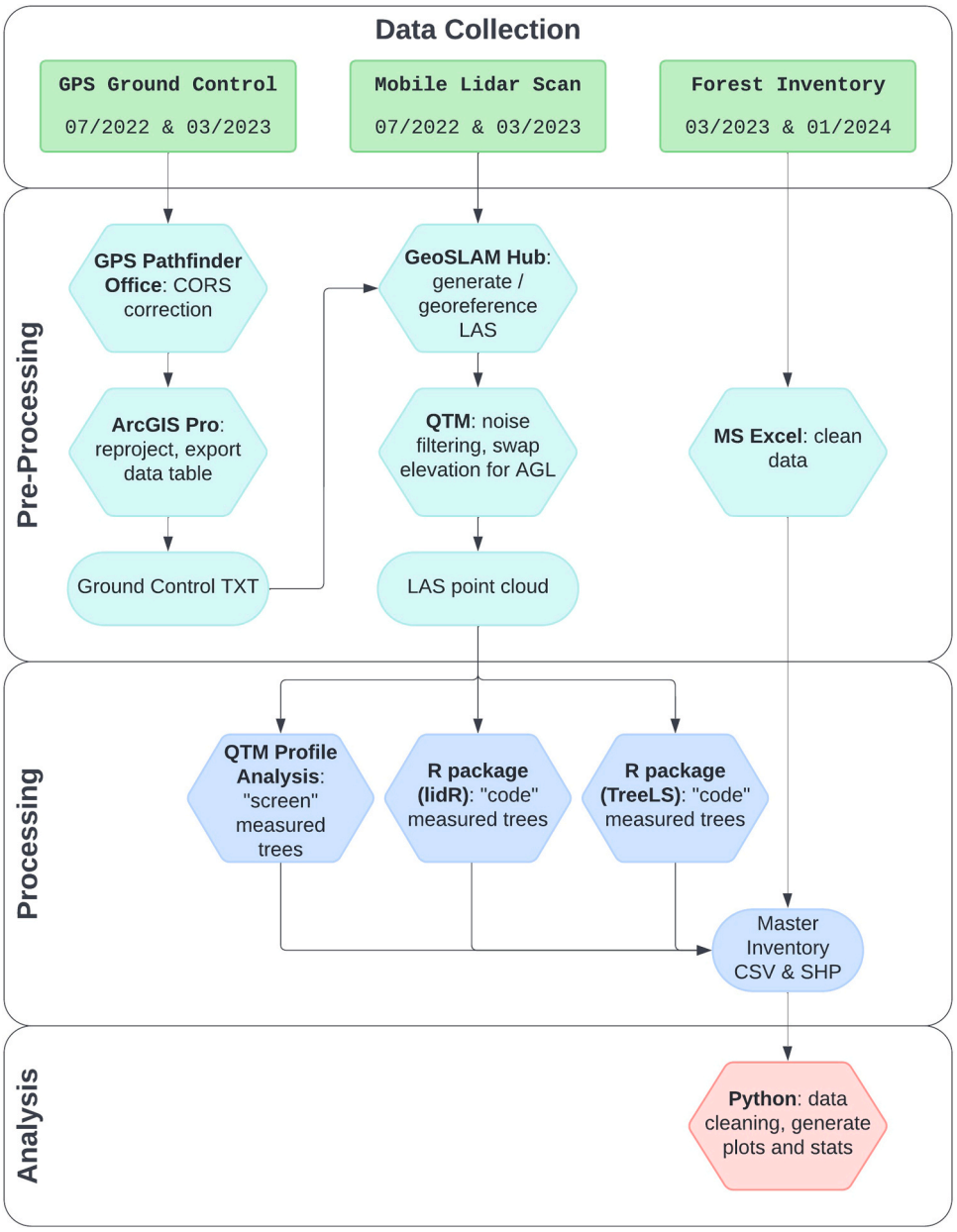


Fig. 15. Flowchart of work from data collection to analysis.

Table 2
Sample size (n), test statistic (s), and p-value (p) for paired *t*-test (PTT) as determined appropriate by a Shapiro-Wilk (S-W) test for normality on the difference of the pairs for all measurements, absolute value of mean difference between pairs (|MD=), Cohen's d effect size (Cd), and interpretation of the PTT and Cd results.

α = 0.01	n	s/p	S-W	PTT	|MD|	Cd	Interpretation
height							
field vs		s	0.947	-4.489			different
screen	50	p	0.025	0.000	0.339 m	-0.635	(-) medium
field vs		s	0.980	-2.624			same (marginal)
lidR	122	p	0.065	0.010	0.208 m	-0.238	(-) small
field vs		s	0.976	-11.337			different
TreeLS	64	p	0.241	0.000	1.600 m	-1.417	(-) very large
dbh							
field vs		s	0.954	11.435			different
screen	50	p	0.048	0.000	5.522 cm	1.617	(+) very large
field vs		s	0.990	2.693			different
TreeLS	64	p	0.867	0.009	1.783 cm	0.337	(+) small

found an $R^2 = 0.91$, RMSE = 5.25 cm, p-value < 0.001, and y = 1.06 x – 4.78.

Table 3
Global Statistic (GS) p-value for each measurement pair.

α = 0.05	GS p-value	Decision
height: field vs screen	0.163	Assumptions acceptable
height: field vs lidR	< 0.001	Assumptions NOT satisfied
height: field vs lidR (above)	0.570	Assumptions acceptable
height: field vs lidR (below)	0.071	Assumptions acceptable
height: field vs TreeLS	0.109	Assumptions acceptable
dbh: field vs screen	0.096	Assumptions acceptable
dbh: field vs TreeLS	0.226	Assumptions acceptable

4. Discussion

4.1. Field Data

The idea that field measurements are always "true" measurements can occasionally be disingenuous. In the case of measuring dbh with a steel diameter tape, field measures are certainly as close to true as possible; however, measuring height using the height routine of a laser rangefinder can be problematic. Through casual observation, we have found that a single tree, measured multiple times from multiple observers, yielded height results with alarming variance. One source of this error is that the highest spot on a tree may be difficult to shoot with a laser rangefinder. Fig. 18 shows a view through the eyepiece of the laser rangefinder used in the study. Notice that the exact top of the tree can be difficult to receive a laser return from, particularly with trees of conical shape, like a young loblolly pine (Pinus taeda), compounded by movement from wind. In such a case, we often "settle" for a lower point on the tree from which we can record an observation. U.S. Forest Service guidelines (USFS, 2022, USFS, 2023) allow for a tolerance of + /- 10 % of true length (tree height) with a measurement quality objective of at least 90 % of the time.

A second source of error is the fact that the highest point of a tree may not actually be observable from the ground, especially in the context of large, mature trees forming dense canopy cover in close proximity to buildings, as is often the case with urban trees. In such an instance, the highest observable point on the tree was recorded. A third source of error can come from the occlusion of the base of the stem, often by dense understory vegetation or shrubbery.

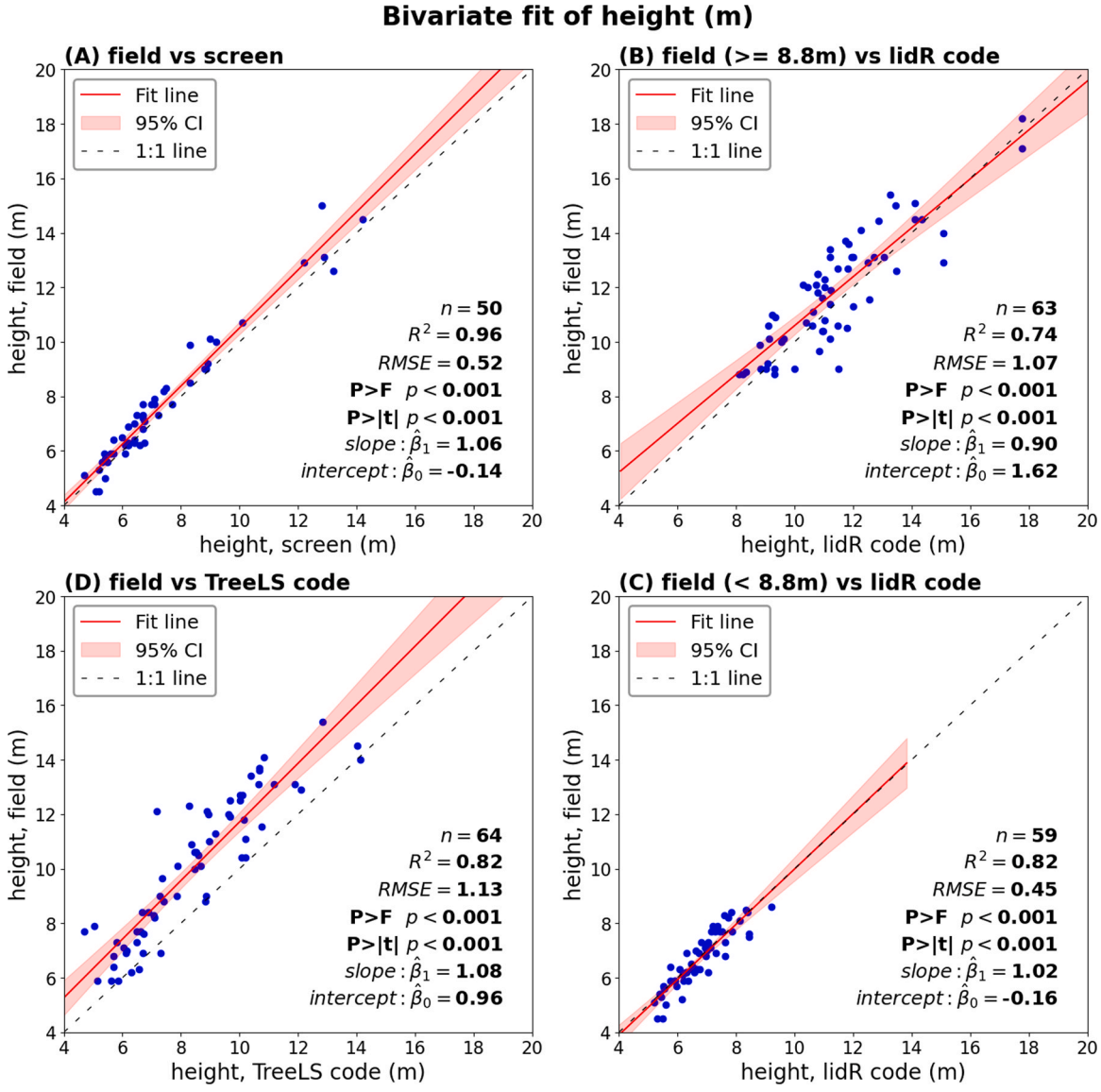


Fig. 16. Bivariate fit of total tree height, comparing (A) field vs screen, (B) field (at or above 8.8 m) vs lidR code, (C) field (below 8.8 m) vs lidR code, and (D) field vs TreeLS code.

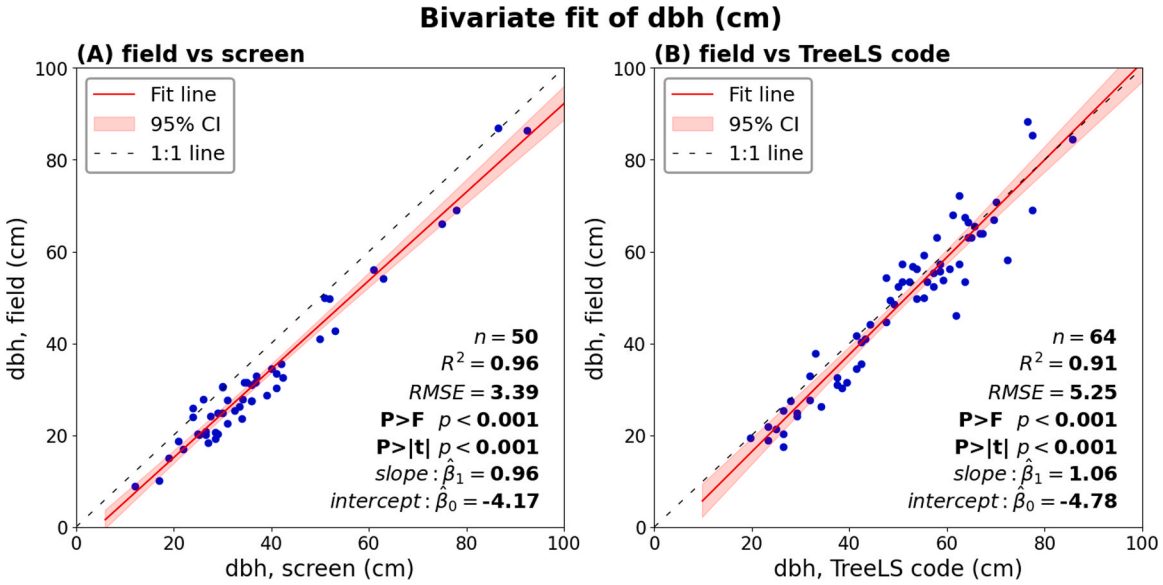


Fig. 17. Bivariate fit of diameters at breast height, comparing (A) field vs screen and (B) field vs TreeLS code.



Fig. 18. Using the height routine of a laser rangefinder to measure tree height; viewed through the eyepiece.

4.2. Statistics

Multiple paired tests were performed on the measurement pairs: three for height and two for dbh. To mitigate an inflated experimentwise Type I error rate (α_E), we opted for a lower Type I error rate for individual comparisons (α_I) by using $\alpha = 0.01$. At this level, the α_E would reach a maximum of 0.030 for three comparisons and 0.020 for two comparisons, well within our acceptable limits.

In comparing heights collected from field and screen (Table 2), the PTT suggested that the pairs were significantly different with a mean difference absolute value of 0.339 m. The Cd yielded a negative medium effect size, which showed that screen measurements tended to be lower than field measurements. In the regression analysis (Fig. 16A), the fit line lies slightly yet fully to the left of the 1:1 line, suggesting that screen heights were under measured with an RMSE of 0.52 m.

Comparing heights from field and lidR code (Table 2), the PTT suggested that the pairs were marginally similar with a 0.208 m mean difference. The Cd yielded a negative small effect size, which indicated that code measurements tended to be slightly lower than field measurements. In regressing the pairs, we found that trees below 8.8 m in height (Fig. 16C) showed lower variance of residuals and closely followed the 1:1 line, while taller trees (Fig. 16B) showed higher variance of residuals and strayed slightly from the 1:1 line. This suggested that while the algorithm was highly precise for shorter / younger trees with simpler crown architectures, the increased canopy complexity, irregular crown morphology, or overlapping canopy layers of taller / mature trees were likely responsible for the tendency to slightly under-measure their total height. When we analyzed the ratio of absolute value of the mean difference between pairs of height measurements (0.208 m) and the mean height of field-sampled trees (9.3 m), we saw that the resulting 2.24 % was well-below the U.S. Forest Service guidelines (USFS, 2022, USFS, 2023) tolerance of + /- 10 % of true length (tree height). The small values of mean difference, effect size, and error tolerance indicated that in terms of ecological significance, lidR was a good surrogate for field measurements.

Comparing heights from field and TreeLS code (Table 2), the PTT suggested that the pairs were significantly different with a 1.600 m mean difference. The Cd yielded a negative very large effect size, which indicated that code measurements were consistently and substantially lower than field measurements. In the regression analysis (Fig. 16D), nearly all points fell to the left of the 1:1 line, which suggested that TreeLS was under-measuring height. This was to be expected, because through visual evaluation of the code plots (Fig. 19), many trees were not measured to the highest point, even with a 0.999 height percentile parameter (99.9th percentile). We hypothesized this to be the case

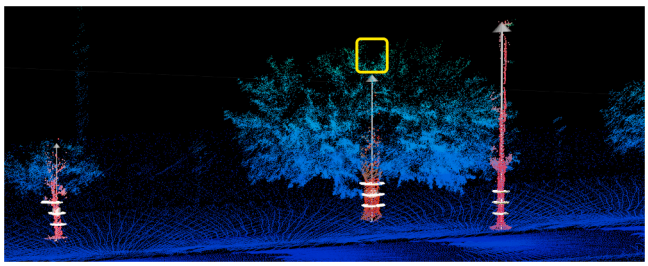


Fig. 19. TreeLS code plot depicting a low height measurement of the taller tree (yellow box) while more correctly measuring height of the shorter tree (left) and streetlamp (right).

because it seemed that only lidar points directly above the stem (within a specified width) were considered for height, so a tree top point out of that range was not considered in height measurement. As a note, a 1.00 height percentile parameter (100th percentile, highest point) tended to include lidar noise points above a tree in the height measurement.

Due to the 100 m range of the scanner used in our study, we were able to satisfactorily reconstruct the tree canopy, compared to mobile lidar scanners used in other studies (Ryding et al., 2015, Cabo et al., 2018, and Chen, S. et al., 2019) The RMSEs that we found for height comparisons (Fig. 16) ranged from 0.45 m (field vs lidR code for shorter trees) to 1.13 m (field vs TreeLS code), lower than those shown by Srinivasan et al. (2015), Saarinen et al. (2017), and Su et al. (2021).

In comparing dbh from field and screen (Table 2), the PTT suggested that the pairs were significantly different with a 5.522 cm mean difference. The Cd yielded a positive very large effect size, which indicated that screen measurements were consistently and substantially larger than field measurements. In the regression analysis (Fig. 17A), nearly all points fell to the right of the 1:1 line, which suggested that dbh was over-measured on screen. This was to be expected due to the high degree of noise inherent with lidar in general, and mobile lidar specifically. As a note, screen measurements for dbh were collected in a manner to reduce the influence of noise as much as possible and without the influence of *a priori* field data.

Comparing dbh from field and TreeLS code (Table 2), the PTT suggested that the pairs were marginally different with a 1.783 cm mean difference. The Cd yielded a positive small effect size, which indicated that code measurements were slightly larger than field measurements. In the regression analysis (Fig. 17B), points below 50 cm tended to fall to the right of the 1:1 line, while points above tended to fall to the left, suggesting that trees of smaller diameter were over-measured by code, while larger trees were under-measured. This was likely the case due to lidar noise having more influence on smaller trees, as it constituted a larger percentage of measured diameter. For this measurement pair, we found an RMSE of 5.25 cm, lower than Chen, Y. et al. (2019), but higher than Ryding et al. (2015), Bauwens et al. (2016), Cabo et al. (2018), Chen, S. et al. (2019), Li et al. (2021), Stal et al. (2021), and Su et al. (2021). We also fully acknowledge that the R^2 of 0.91 only represents the trees that the algorithm actually detected and that these values suffer from a survivorship bias.

4.3. Sample sizes

Initially, 50 trees were field sampled as a pilot study; an additional 72 trees were added for a total of 122 trees. For each of these initial 50 field trees, measurements of height, dbh, and crown width were collected on screen as a proof of concept before code evaluation. The lidR code identified and measured all field-sampled trees. Across the entire study area, TreeLS code was only able to identify 64 field-sampled trees. We hypothesized that this was for three reasons. First, the package identifies a tree by searching for circles in the point cloud. If a tree is too far from the scan line and there are too few points on target, then circles cannot be found, and trees cannot be identified. Of the trees located, the distance to scan line ranged from 0.8 m – 18.5 m, with a mean of 5.3 m and median of 4.1 m.

Second, an idiosyncrasy of the package was that urban artifacts (e.g., streetlamps and road signs) in close proximity to trees tended to "confuse" the code, and a bogus location somewhere between the two objects was erroneously chosen. An example of this phenomenon is shown in Fig. 20. Notice that the 'treeMap' function (yellow pancakes) correctly identified each tree, in addition to both streetlamps, but this did not translate to correctly identifying 'treePoints' and 'stemPoints'. The alternative to this issue would have been to remove these problematic urban artifacts from the point cloud, but until this process can be automated, it remains tediously manual.

Third, the parameters for each function in the process were optimized on a sample subset of 10 trees from the full dataset. However,

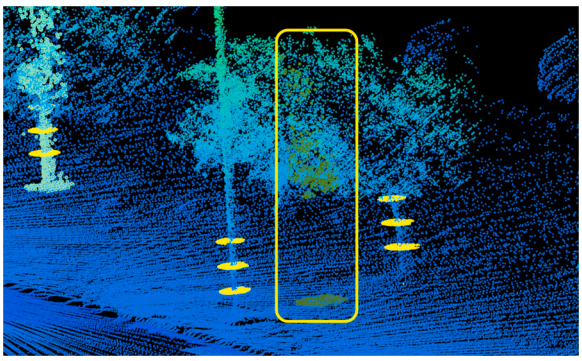


Fig. 20. TreeLS code misidentified a tree (yellow box). Notice that other trees and the streetlamp were identified by the 'treeMap' function (yellow pancakes).

when this process was attempted on the full MLS point cloud, the script stalled on step one, even after running a point thinning process. The specific nature of this fatal stall was never discovered as the package failed to provide any useful message for status or error. After an unreasonable amount of time with no progress, it was decided to manually subset the MLS dataset into smaller pieces, eventually settling on 16 subsets. The subset process was completed using the clip tool in QTM and was performed carefully to not disturb individual trees. Other quantities and sizes of subsets were not evaluated. It is of note that each subset ran, start to finish, in under 45 s, suggesting that this was not a computational resource limitation. Future improvements to TreeLS should eliminate the necessity for such manual workarounds and lead to smoother automation. Each subset process resulted in a CSV of tree locations and measurements for height and dbh. All 16 datasets were combined, yielding a single dataset. When these locations were spatially joined to the master inventory, only 64 TreeLS locations remained. Upon further investigation, we learned that the script failed to locate even some of the initial "training trees" from the subset (Fig. 21). The reason for this is unknown but seems to be the third source of error in measuring trees. Overall, the missing trees seemed to be a combination of various factors, including scan path distance and algorithm limitations. While the observed high omission rate (>47 %) of detected trees limits the package's viability as a standalone tool for automated inventory, we expect that future versions may mitigate this issue. Additional factors such as occlusion and canopy density were not specifically investigated.

4.4. IMU and data drift

The scanner has an onboard inertial measurement unit (IMU) that records the motion of the scanner to reconstruct the lidar point cloud. In testing, we found this IMU to be sensitive to certain types of motion, such as driving over a pothole in the road, which could lead to data drift. Care must be taken to reduce jarring as much as possible. Future studies will employ a stabilizing gimbal to aid in this effort.

4.5. Open-source tools, proprietary software, and full automation

While we prioritized the use of open-source tools wherever feasible,

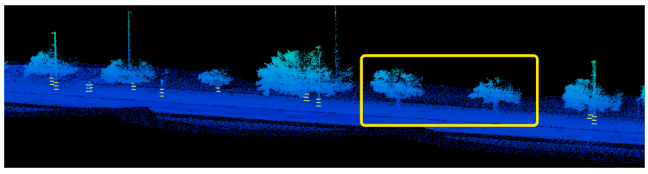


Fig. 21. TreeLS failed to identify and measure some trees (yellow box), yet still located streetlamps and road signs.

portions of the current pre-processing and processing workflow rely on proprietary software, highlighting opportunities for future open-source development and integration. Some open-source tools such as CloudCompare (CloudCompare, 2024) or QGIS (QGIS Development Team, 2024) could be leveraged as alternatives to QTM and ArcGIS Pro; however, GeoSLAM Hub is still required if using GeoSLAM scanners. Furthermore, we acknowledge that many of these pre-processing and processing steps currently serve as hurdles or bottlenecks to fully automating the urban tree inventory process.

4.6. Limitations

For the purposes of our investigations, we specifically chose to field-sample a large number of well-formed trees within a managed and manicured campus environment to serve as a best-case scenario for workflow development. As such, no additional study locations were deemed necessary. We understand that these choices somewhat limit the immediate adaptation of our methodologies to more natural environments or different scales and scopes. Furthermore, we did not specifically address certain limitations of mobile lidar, such as tree occlusion in dense canopies or mixed-species detection accuracy. Time and computational efficiencies were not specifically investigated as these have already been studied and reported by Chen, S. et al. (2019), Ryding et al. (2015), and Lindenbergh et al. (2015). We did not specifically investigate or troubleshoot the various "failures" of the open-source packages, rather, we documented the circumstances and reported the workarounds as appropriate. As is, the high omission rate of detected trees by the TreeLS package makes it unsuitable for automated tree inventory, but we expect that future releases and other packages will close this gap. Similarly, we did not perform a spatial analysis of the detection omissions.

4.7. Future directions

To investigate further, future studies could 1) investigate a more diverse assemblage of tree species with complex canopy structures in various environmental conditions, 2) scale-up these analyses to areas inaccessible to mobile lidar scanning with the inclusion of airborne lidar scanning, 3) investigate various methods of point cloud noise removal automation and the effects on measurement estimations, 4) quantify cost savings in labor of utilizing mobile lidar scanning compared to traditional field measurements in an urban environment, a naturally regenerated forest, and in orchard production, 5) estimate the economic impact of trade in carbon credits, and 6) model and quantify shade of urban trees and how it relates to urban heat island effect.

Additionally, future research should incorporate a spatial analysis of detection omission to quantify how various factors affect algorithmic reliability, including tree diameter, distance from scan line, and proximity to urban features such as buildings, streetlamps, road signs, and competing vegetation. Such an investigation would serve as a strong diagnostic tool for developing and improving forest-centric software for use in a complex urban environment.

In this study, we demonstrated how on-screen mobile lidar measurements served as an intermediate validation step between field data collection and algorithmic processing. We suggest that future research should adopt a similar, multi-tiered, validation framework which explicitly separates data acquisition performance from downstream analytical performance, allowing hardware fidelity to be assessed independently from software implementation. We offer this "field vs screen" validation strategy and recommend it to be adopted as a best practice for future lidar investigations, particularly when establishing or benchmarking novel processing workflows.

To make our work more accessible and actionable to practitioners and policymakers, we offer the considerations below for integration within municipal urban forestry departments. We emphasized the use of open-source platforms, such as R and Python, but there are also cost / benefit considerations for purchased and subscription-based software, such as Quick Terrain Modeler and ArcGIS Pro. While the mobile lidar equipment is relatively inexpensive, there is a learning curve to proper data collection and processing. Although our data processing was performed on a high-end workstation, a much lower-cost PC with modest resources could be used, especially if working in small batches. Long-term data storage and data security measures will need to be addressed as needed. Many of the data processing outputs in our workflow are shapefiles and tabular data, which are easily integrated into existing geospatial protocols. These software, equipment, and personnel costs can be leveraged by "sharing" with Utilities, Engineering, GIS, and IT Departments.

5. Conclusions

Urban trees are an important component of the built environment, providing numerous environmental benefits. In order to more fully understand the importance of the various ecosystem services provided by urban trees, we must be able to map, identify, and measure individual trees as accurately and efficiently as possible. Automating this approach using lidar and aerial photography can improve efficiencies and affordability of urban tree inventories. Beyond basic tree biometrics, the identification and measurement of urban trees is critical for effective urban planning, the detection of hazardous trees, and the evaluation of their role in climate regulation, carbon sequestration, and water cycling. Municipalities and urban planners could utilize mobile lidar data for periodic monitoring, risk assessment, and the development of climate adaptation strategies.

To our knowledge, handheld / backpack-mounted mobile lidar scanning had not been collected or analyzed in our same manner as a tool for urban tree ecology. We developed a complete and accessible workflow for contemporary urban forestry. We collected mobile lidar data with a 100 m range scanner, resulting in accurate representations of tree canopies. We emphasized the importance of collecting mobile lidar scans 1) via scooter rather than by vehicle and 2) concurrent to GPS reference data collection. This electric scooter workflow was a targeted solution to the middle-scale gap in urban forestry: areas too extensive for walking surveys yet inaccessible to vehicle-mounted surveys. We developed the methodology of using on-screen mobile lidar measurements to 1) serve as a robust, high-fidelity validation tier and 2) act as a surrogate for field data to develop and expand parameter extraction algorithms. We developed the Synthetic Point from Raster tool as a novel approach for lidar point thinning, resulting in significantly smaller datasets and shorter computational times. This adaptable solution directly addresses a "big data" problem in mobile lidar, with applications far beyond urban forestry. We explored and discussed algorithmic behavior on various age classes of trees, leading to a greater understanding and implementation of these processes. We identified algorithmic processing challenges and limitations, providing a practical assessment of the performance of forestry-centric tools in a complex urban environment. We employed traditional forest inventory methods to measure 122 urban trees for use as field validation. The integrated workflow system, the rigor of multi-tiered validation, and the insights gained from algorithmic challenges were presented as significant methodological contributions. Our investigation was part of a growing movement toward semi-automation and scalability in urban forestry and urban remote sensing. We were able to demonstrate that mobile lidar data collection is an accurate and viable alternative to traditional forest inventory methods.

Funding

This work was supported by the Carder Graduate Assistantship and the Vola and Dr. Fred '59 Palmer Scholarship.

CRediT authorship contribution statement

Sorin C. Popescu: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Justinn J. Jones:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Stephen L. Webb:** Writing – review & editing. **Joshuah S. Perkin:** Writing – review & editing, Software, Resources, Formal analysis.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Microsoft Copilot and Google Gemini to locate and examine relevant articles, refine R and Python code, and improve efficiencies in the writing process. After using this tool / service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supervised by a thesis committee consisting of Professor Sorin C. Popescu [advisor] and Associate Professor Joshuah S. Perkin of the Department of Ecology and Conservation Biology, and Research Assistant Professor Stephen L. Webb of the Department of Rangeland, Wildlife, and Fisheries Management. All work conducted for the thesis was completed by the student independently. Graduate study was supported financially by the Carder Graduate Assistantship and the Vola and Dr. Fred '59 Palmer Scholarship.

Nomenclatures

3DEP	3D Elevation Program
AGL	Above ground level
AGP	ArcGIS Pro (software)
ALS	Airborne laser / lidar scanning
CHM	Canopy height model
CORS	Continuously Operating Reference Station
CSV	Comma-separated values file format (.csv)
CW	Crown width
DBH	Diameter at breast height
DEM	Digital elevation model
DTM	Digital terrain model
GNSS	Global navigation satellite system
GPS	Global positioning system
H	Height, total tree
ICW	Improved crown width
IMU	Inertial measurement unit
IPYNB	Interactive python notebook
LAS	LASer file format (.las)
LAZ	Zipped LAS file format (.laz)
MLS	Mobile laser / lidar scanning
MSL	Mean sea level
NAIP	National Agriculture Imagery Program
NDVI	Normalized difference vegetation index
PPM	Points per meter squared
PTT	Paired *t*-test
QTM	Quick Terrain Modeler (software)
R/G/B/NIR	Red, green, blue, near-infrared
R^2 / R^2	Coefficient of determination
RGB	Red, green, blue
RMSE	Root mean squared error
SHP	ESRI shapefile format (.shp)
SPR	Synthetic Points from Raster tool (python notebook)
S-W	Shapiro-Wilk
TLS	Terrestrial laser / lidar scanning
TOC	Top of canopy
TXT	Text file format (.txt)
WSR	Wilcoxon signed-rank test
XLS	Microsoft Excel file (.xls /.xlsx)

References

Alvites, C., Marchetti, M., Lasserre, B., Santopuoli, G., 2022. LiDAR as a Tool for Assessing Timber Assortments: A Systematic Literature Review. Remote Sens. 14 (18), 4466. https://doi.org/10.3390/rs14184466.

Appelhans, T., Detsch, F., Reudenbach, C., Woellauer, S., 2023. mapview: Interactive Viewing of Spatial Data in R. R package version 2.11.2. ⟨https://cran.r-project.org/web/packages/mapview/index.html⟩.

Applied Imagery, 2024. Quick Terrain Modeler (v8.2.3). ⟨https://appliedimagery.com⟩.

Aschoff, T., Spiecker, H., 2004. Algorithms for the automatic detection of trees in laser scanner data. Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci. 36, 66–70.

Bauwens, S., Bartholomeus, H., Calders, K., Lejeune, P., 2016. Forest Inventory with Terrestrial LiDAR: A Comparison of Static and Hand-Held Mobile Laser Scanning. Forests 7, 127. https://doi.org/10.3390/f7060127.

Bienert, A., Scheller, S., Keane, E., Mullooly, G., Mohan, F., 2006. Application of terrestrial laser scanners for the determination of forest inventory parameters. Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci. 36, 5.

Brolly, G., Kiraly, G., 2009. Algorithms for stem mapping by means of terrestrial laser scanning. Acta Silv. Lignaria Hung. 5, 119–130.

Cabo, C., Del Pozo, S., Rodríguez-Gonzálvez, P., Ordóñez, C., González-Aguilera, D., 2018. Comparing Terrestrial Laser Scanning (TLS) and Wearable Laser Scanning (WLS) for Individual Tree Modeling at Plot Level. Remote Sens. 10, 540. https://doi.org/10.3390/rs10040540.

Chen, S., Liu, H., Feng, Z., Shen, C., Chen, P., 2019. Applicability of personal laser scanning in forestry inventory. PLoS ONE 14, e0211392. https://doi.org/10.1371/journal.pone.0211392.

Chen, Y., Wang, S., Li, J., Ma, L., Wu, R., Luo, Z., Wang, X., 2019. Rapid Urban Roadside Tree Inventory Using a Mobile Laser Scanning System. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 12, 3690–3700. https://doi.org/10.1109/JSTARS.2019.2929546.

CloudCompare, 2024. Retrieved from ⟨http://www.cloudcompare.org/⟩.

Côté, J.F., Fournier, R.A., Luther, J.E., van Lier, O.R., 2018. Fine-scale three-dimensional modeling of boreal forest plots to improve forest characterization with remote sensing. Remote Sens. Environ. 219, 99–114. https://doi.org/10.1016/j.rse.2018.09.026.

Dalponte, M., Coomes, D.A., 2016. Tree-centric mapping of forest carbon density from airborne laser scanning and hyperspectral data. Methods Ecol. Evol. 7, 1236–1245. https://doi.org/10.1111/2041-210X.12575.

De Conto, T., 2020. TreeLS: Terrestrial Point Cloud Processing of Forest Data (v2.0.2). ⟨https://github.com/tiagodc/TreeLS/⟩.

Environmental Systems Research Institute (ESRI), 2024. ArcGIS Pro (v3.1.0). ⟨https://www.esri.com⟩.

Fan, W., Chenglu, W., Jonathan, L., 2016. Automated extraction of urban trees from mobile LiDAR point clouds. 2nd ISPRS Int. Conf. Comput. Vis. Remote Sens. (CVRS 2015) 9901 159–164.

FARO, 2024. GeoSLAM Hub (v6.1.0). ⟨https://knowledge.faro.com⟩.

Forsman, M., Holmgren, J., Olofsson, K., 2016. Tree Stem Diameter Estimation from Mobile Laser Scanning Using Line-Wise Intensity-Based Clustering. Forests 7, 206. https://doi.org/10.3390/f7090206.

Golombek, Y., Marshall, W., 2021. High-density mobile LiDAR for measuring urban streetscape features. Urban Des. Int. 27, 3–21. https://doi.org/10.1057/s41289-021-00176-4.

Harris, R, C., et al., 2020. Array programming with NumPy. Nature 585, 357–362. https://doi.org/10.1038/s41586-020-2649-2.

Heo, H.K., Lee, D.K., Park, J.H., Thorne, J.H., 2019. Estimating the heights and diameters at breast height of trees in an urban park and along a street using mobile LiDAR. Landsc. Ecol. Eng. 15, 253–263. https://doi.org/10.1007/s11355-019-00379-6.

Herrero-Huerta, M., Lindenbergh, R., Rodríguez-Gonzálvez, P., 2018. Automatic tree parameter extraction by a Mobile LiDAR System in an urban context. PLOS ONE 13 (4), e0196004. https://doi.org/10.1371/journal.pone.0196004.

Hijmans, R., 2023. raster: Geographic Data Analysis and Modeling. R package version 3.6-26. ⟨https://cran.r-project.org/web/packages/raster/index.html⟩.

Hijmans, R., 2024. terra: Spatial Data Analysis. R package version 1.7-78. ⟨https://cran.r-project.org/web/packages/terra/index.html⟩.

Holopainen, M., Kankare, V., Vastaranta, M., Liang, X., Lin, Y., Vaaja, M., Liang, X., Kukko, A., Kaartinen, H., Hyyppa, J., Hyyppa, H., 2013. Tree mapping using airborne, terrestrial and mobile laser scanning – A case study in a heterogeneous urban forest. Urban For. Urban Green. 12, 546–553. https://doi.org/10.1016/j.ufug.2013.06.002.

Hopkinson, C., Chasmer, L., Young-Pow, C., Treitz, P., 2004. Assessing forest metrics with a ground-based scanning LiDAR. Can. J. For. Res. 34, 573–583. https://doi.org/10.1139/x03-225.

Huang, S., Hager, S.A., Halligan, K.Q., Fairweather, I.S., Swanson, A.K., Crabtree, R.L., 2009. A comparison of individual tree and forest plot height derived from LiDAR and InSAR. Photogramm. Eng. Remote Sens. 75, 159–167. https://doi.org/10.14358/PERS.75.2.159.

Hunter, J.D., 2007. Matplotlib: A 2D Graphics Environment. Comput. Sci. Eng. 9, 90–95. https://doi.org/10.1109/MCSE.2007.55.

Husain, A., Vaishya, R.C., 2019. Detection and thinning of street trees for calculation of morphological parameters using mobile laser scanner data. Remote Sens. Appl. Soc. Environ. 13, 375–388. https://doi.org/10.1016/j.rsase.2018.12.007.

Li, D., Guo, H., Jia, W., Wang, F., 2021. Analysis of Taper Functions for Larix olgensis Using Mixed Models and TLS. Forests 12, 196. https://doi.org/10.3390/f12020196.

Li, Q., Yuan, P., Liu, X., Zhou, H., 2020. Street tree segmentation from mobile laser scanning data. Int. J. Remote Sens. 41, 7145–7162. https://doi.org/10.1080/01431161.2020.1754495.

Lindenbergh, R.C., Berthold, D., Sirmacek, B., Herrero-Huerta, M., Wang, J., Ebersbach, D., 2015. Automated large scale parameter extraction of road-side trees sampled by a laser mobile mapping system. Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci. XL-3/W3, 589–594. https://doi.org/10.5194/isprsarchives-XL-3-W3-589-2015.

Liu, L., Coops, N.C., Aven, N.W., Pang, Y., 2017. Mapping urban tree species using integrated airborne hyperspectral and LiDAR remote sensing data. Remote Sens. Environ. 200, 170–182. https://doi.org/10.1016/j.rse.2017.08.010.

McKinney, W., 2010. Data Structures for Statistical Computing in Python. Proc. 9th Python Sci. Conf. 51–56.

Monnier, F., Vallet, B., Soheilian, B., 2012. Trees detection from laser point clouds acquired in dense urban areas by a mobile mapping system. ISPRS Ann. Photogramm. Remote Sens. Spat. Inf. Sci. I-3, 245–250. https://doi.org/10.5194/isprsannals-I-3-245-2012.

Montoya, O., Icasio-Hernández, O., Salas, J., 2021. TreeTool: A tool for detecting trees and estimating their DBH using forest point clouds. SoftwareX 16, 100867. https://doi.org/10.1016/j.softx.2021.100889.

Moorthy, I., Miller, J.R., Berni, J.A.J., Zarco-Tejada, P., Hu, B., Chen, J., 2011. Field characterization of olive (Olea europaea L.) tree crown architecture using terrestrial laser scanning data. Agric. For. Meteorol. 151, 204–214. https://doi.org/10.1016/j.agrformet.2010.10.005.

Moreira, B.M., Goyanes, G., Pina, P., Vassilev, O., Heleno, S., 2021. Assessment of the Influence of Survey Design and Processing Choices on the Accuracy of Tree Diameter at Breast Height (DBH) Measurements Using UAV-Based Photogrammetry. Drones 5, 126. https://doi.org/10.3390/drones5020043.

Nowak, D.J., 2006. Institutionalizing urban forestry as a "biotechnology" to improve environmental quality. Urban For. Urban Green. 5, 93–100. https://doi.org/10.1016/j.ufug.2006.04.002.

Pebesma, E., 2018. Simple Features for R: Standardized Support for Spatial Vector Data. R. J. 10 (1), 439–446. https://doi.org/10.32614/RJ-2018-009.

Pebesma, E., Bivand, R., 2023. Spatial Data Science: With Applications in R. Chapman and Hall/CRC. https://doi.org/10.1201/9780429459016.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E., 2011. Scikit-learn: Machine learning in Python. J. Mach. Learn. Res. 12, 2825–2830. ⟨http://jmlr.org/papers/v12/pedregosa11a.html⟩.

Peña, E.A., Slate, E.H., 2006. Global Validation of Linear Model Assumptions. J. Am. Stat. Assoc. 101, 341–354. https://doi.org/10.1198/016214505000000637.

Popescu, S.C., Wynne, R.H., Nelson, R.F., 2003. Measuring individual tree crown diameter with lidar and assessing its influence on estimating forest volume and biomass. Can. J. Remote Sens. 29, 564–577. https://doi.org/10.5589/m03-027.

Popescu, S.C., Wynne, R.H., 2004. Seeing the trees in the forest: Using lidar and multispectral data fusion with local filtering and variable window size for estimating tree height. Photogramm. Eng. Remote Sens. 70, 589–604. https://doi.org/10.14358/PERS.70.5.589.

Popescu, S.C., Wynne, R.H., Scrivani, J.A., 2004. Fusion of small-footprint lidar and multispectral data to estimate plot-level volume and biomass in deciduous and pine forests in Virginia, USA. For. Sci. 50, 551–565. https://doi.org/10.1093/forestscience/50.4.551.

Popescu, S.C., 2007. Estimating biomass of individual pine trees using airborne lidar. Biomass.. Bioenergy 31, 646–655. https://doi.org/10.1016/j.biombioe.2007.06.022.

Popescu, S.C., Zhao, K., 2008. A voxel-based lidar method for estimating crown base height for deciduous and pine trees. Remote Sens. Environ. 112, 767–781. https://doi.org/10.1016/j.rse.2007.06.011.

Putman, E.B., Popescu, S.C., 2018. Automated Estimation of Standing Dead Tree Volume Using Voxelized Terrestrial Lidar Data. IEEE Trans. Geosci. Remote Sens. 56, 6484–6503. https://doi.org/10.1109/TGRS.2018.2839088.

Puttonen, E., Jaakkola, A., Litkey, P., Hyyppä, J., 2011. Tree Classification with Fused Mobile Laser Scanning and Hyperspectral Data. Sensors 11, 5158–5182. https://doi.org/10.3390/s110505158.

Python Software Foundation, 2023. Python Language Reference (v3.9.18). ⟨https://www.python.org/downloads/⟩.

QGIS Development Team, 2024. QGIS Geographic Information System (Version 3.34). Open Source Geospatial Foundation Project. ⟨http://qgis.org⟩.

R Core Team, 2023. R: A language and environment for statistical computing (v4.3.1). ⟨https://www.R-project.org/⟩.

Roussel, J., Auty, D., Coops, N.C., Tompalski, P., Goodbody, T.R.H., Meador, A.S., Bourdon, J.F., de Boissieu, F., Brochu, A., Béland, M., Remmel, T.K., Havrelock, C., 2020. lidR: An R package for analysis of Airborne Laser Scanning (ALS) data. Remote Sens. Environ. 251, 112061. https://doi.org/10.1016/j.rse.2020.112061.

Roussel, J., Auty, D., 2024. lidR: Airborne LiDAR Data Manipulation and Visualization for Forestry Applications. R package version 4.1.1. ⟨https://cran.r-project.org/package=lidR⟩.

RStudio Team, 2024. RStudio: Integrated Development for R (v2023.12.0). ⟨https://posit.co/downloads/⟩.

Rutzinger, M., Pratihast, A.K., Oude Elberink, S., Vosselman, G., 2010. Detection and modelling of 3D trees from mobile laser scanning data. Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci. 38, 520–525.

Ryding, J., Williams, E., Smith, M., Eichhorn, M., 2015. Assessing Handheld Mobile Laser Scanners for Forest Surveys. Remote Sens. 7, 1095–1111. https://doi.org/10.3390/rs70101095.

Saarinen, N., Kankare, V., Vastaranta, M., Luoma, V., Pyörälä, J., Tanhuanpää, T., Liang, X., Kaartinen, H., Kukko, A., Jaakkola, A., Holopainen, M., Hyyppä, J., Vastaranta, M., 2017. Feasibility of Terrestrial laser scanning for collecting stem volume information from single trees. ISPRS J. Photogramm. Remote Sens. 123, 140–158. https://doi.org/10.1016/j.isprsjprs.2016.11.012.

Safaie, A.H., Rastiveis, H., Shams, A., Sarasua, W.A., Li, J., 2021. Automated street tree inventory using mobile LiDAR point clouds based on Hough transform and active contours. ISPRS J. Photogramm. Remote Sens. 174, 19–34. https://doi.org/10.1016/j.isprsjprs.2021.01.026.

Seabold, S., Perktold, J., 2010. statsmodels: Econometric and statistical modeling with python. Proc. 9th Python Sci. Conf.

Shrestha, R., Wynne, R.H., 2012. Estimating Biophysical Parameters of Individual Trees in an Urban Environment Using Small Footprint Discrete-Return Imaging Lidar. Remote Sens. 4, 484–508. https://doi.org/10.3390/rs4020484.

Sirmacek, B., Lindenbergh, R., 2015. Automatic Classification Of Trees From Laser Scanning Point Clouds. ISPRS Ann. Photogramm. Remote Sens. Spat. Inf. Sci. II3/W4 217224. https://doi.org/10.5194/isprsannals-II-3-W5-137-2015.

Srinivasan, S., Popescu, S., Eriksson, M., Sheridan, R., Ku, N., 2015. Terrestrial Laser Scanning as an Effective Tool to Retrieve Tree Level Height, Crown Width, and Stem Diameter. Remote Sens. 7, 1877–1896. https://doi.org/10.3390/rs70201877.

Stal, C., Verbeurgt, J., De Sloover, L., De Wulf, A., 2021. Assessment of handheld mobile terrestrial laser scanning for estimating tree parameters. J. For. Res. 32, 1503–1513. https://doi.org/10.1007/s11676-020-01214-7.

Su, Y., Guo, Q., Jin, S., Guan, H., Sun, X., Ma, Q., Hu, T., Wang, R., Li, Y., 2021. The Development and Evaluation of a Backpack LiDAR System for Accurate and Efficient Forest Inventory. IEEE Geosci. Remote Sens. Lett. 18, 1660–1664. https://doi.org/10.1109/LGRS.2020.3005166.

Tansey, K., Selmes, N., Anstee, A., Tate, N.J., Denniss, A., 2009. Estimating tree and stand variables in a Corsican Pine woodland from terrestrial laser scanner data. Int. J. Remote Sens. 30, 5195–5209. https://doi.org/10.1080/01431160902882587.

Thies, M., Spiecker, H., 2004. Evaluation and future prospects of terrestrial laser scanning for standardized forest inventories. Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci. 36, 192–197.

Trimble, 2013. GPS Pathfinder Office (v5.60). ⟨https://forms.trimble.com/globalTRLTAB.aspx?Nav=Collection-4552⟩.

U.S. Dept. of Agriculture (USDA), 2023. Natl. Agric. Imag. Program. ⟨https://naip-usdaonline.hub.arcgis.com/⟩.

U.S. Forest Service (USFS), 2022. Forest Inventory and Analysis National Urban FIA Plot Field Guide Field Data Collection Procedures for Urban FIA Plots (version 9.2). ⟨https://www.fs.usda.gov/research/programs/urbanfia#resources⟩.

U.S. Forest Service (USFS), 2023. Forest Inventory and Analysis National Core Field Guide for the Nationwide Forest Inventory (version 9.3). ⟨https://www.fs.usda.gov/research/understory/nationwide-forest-inventory-field-guide⟩.

U.S. Geological Survey (USGS), 2023. USGS 3D Elevation Program. ⟨https://www.usgs.gov/3d-elevation-program⟩.

Van Rossum, G., 2020. The Python Library Reference, release 3.9.18. Python Softw. Found. ⟨https://www.python.org/downloads/⟩.

Velasquez-Camacho, L., Cardil, A., Mohan, M., Etxegarai, M., Anzaldi, G., De-Miguel, S., 2021. Remotely Sensed Tree Characterization in Urban Areas: A Review. Remote Sens. 13, 2627. https://doi.org/10.3390/rs13234889.

Virtanen, P., Gommers, R., Oliphant, T.E., et al., 2020. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. Nat. Methods 17 (3), 261–272. https://doi.org/10.1038/s41592-019-0686-2.

Visual Studio Code, 2023. Visual Studio Code (v1.89.1). ⟨https://code.visualstudio.com/Download⟩.

Wang, K., Wang, T., Liu, X., 2018. A Review: Individual Tree Species Classification Using Integrated Airborne LiDAR and Optical Imagery with a Focus on the Urban Environment. Forests 10, 6. https://doi.org/10.3390/f10010001.

Watt, P.J., Donoghue, D.N.M., 2005. Measuring forest structure with terrestrial laser scanning. Int. J. Remote Sens. 26, 1437–1446. https://doi.org/10.1080/01431160512331337961.

Wezyk, P., Koziol, K., Glista, M., Pierzchalski, M., 2007. Terrestrial laser scanning versus traditional forest inventory. First results from the Polish forests. Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci. 36, 424–429.

Wich, S.A., Piel, A.K., 2021. Oxford. Conservation technology. Oxford University Press.

Wickham, H., 2016. Data Analysis. In: ggplot2. Use R!. Springer, Cham. https://doi.org/10.1007/978-3-319-24277-4_9.

Windrim, L., Bryson, M., 2020. Detection, Segmentation, and Model Fitting of Individual Tree Stems from Airborne Laser Scanning of Forests Using Deep Learning. Remote Sens. 12, 1469. https://doi.org/10.3390/rs12091469.

Wood, J.P., 1999. Tree inventories and GIS in urban forestry (Master's project report). Virginia Polytechnic Institute and State University, Blacksburg, Virginia.

Wu, B., Yu, B., Yue, W., Shu, S., Tan, W., Hu, C., et al., 2013. A Voxel-Based Method for Automated Identification and Morphological Parameters Estimation of Individual Street Trees from Mobile Laser Scanning Data. Remote Sens. 5, 584–611. https://doi.org/10.3390/rs5020584.

Wu, J., Yao, W., Polewski, P., 2018. Mapping Individual Tree Species and Vitality along Urban Road Corridors with LiDAR and Imaging Sensors: Point Density versus View Perspective. Remote Sens. 10, 1403. https://doi.org/10.3390/rs10091403.

Xu, S., Xu, S., Ye, N., Zhu, F., 2018. Automatic extraction of street trees' nonphotosynthetic components from MLS data. Int. J. Appl. Earth Obs. Geoinf. 69, 64–77. https://doi.org/10.1016/j.jag.2018.02.016.

Yan, Z., Liu, R., Cheng, L., Zhou, X., Ruan, X., Xiao, Y., 2019. A Concave Hull Methodology for Calculating the Crown Volume of Individual Trees Based on Vehicle-Borne LiDAR Data. Remote Sens. 11, 623. https://doi.org/10.3390/rs11060623.

Yao, T., Yang, X., Zhao, F., Wang, Z., Zhang, Q., Jupp, D., Lovell, J., Culvenor, D., Newnham, G., Ni-Meister, W., et al., 2011. Measuring forest structure and biomass in New England forest stands using Echidna ground-based lidar. Remote Sens. Environ. 115, 2965–2974. https://doi.org/10.1016/j.rse.2010.03.019.

Zhang, C., Zhou, Y., Qiu, F., 2015. Individual tree segmentation from LiDAR point clouds for urban forest inventory. Remote Sens. 7, 7892–7913. https://doi.org/10.3390/rs70607892.

Zhong, L., Cheng, L., Xu, H., Wu, Y., Chen, Y., Li, M., 2017. Segmentation of Individual Trees from TLS and MLS Data. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 10, 774–787. https://doi.org/10.1109/JSTARS.2016.2565519.

Zhong, R., Wei, J., Su, W., Chen, Y.F., 2013. A method for extracting trees from vehicle-borne laser scanning data. Math. Comput. Model. 58, 727–736. https://doi.org/10.1016/j.mcm.2012.12.038.